

C. F. Gauss
Theory of Biquadratic Residues. Second Memoir
Articles 30–76

Art. 30

Induction therefore opens an exceedingly rich harvest of special theorems akin to the theorem for the number 2; but the common bond is lacking, and rigorous demonstrations are lacking, since the method by which, in the first memoir, we settled the number 2 does not admit further application. There are indeed other methods by which one could obtain demonstrations for particular cases, especially those concerning the distribution of quadratic residues among the complexes A and C ; but we shall not dwell on them, since what is wanted is a general theory embracing all cases.

When, from the year 1805 onward, we began to devote our reflections to this matter, we soon became certain that the genuine source of the general theory must be sought in a promoted, or extended, field of arithmetic, as we already indicated in Article 1.

Just as higher arithmetic, in the questions treated up to this point, ranges only over real integers, so the theorems concerning biquadratic residues shine forth in their full simplicity and genuine beauty only when the field of arithmetic is extended to imaginary quantities, so that, without restriction, its objects are the numbers of the form $a + bi$, where i denotes, as usual, the imaginary quantity $\sqrt{-1}$, and a and b range indefinitely over all real integers from $-\infty$ to $+\infty$. We shall call such numbers *complex integral numbers*, in such a way that real numbers are not opposed to complex numbers, but are instead regarded as a species included among them. The present memoir will set forth both the elementary doctrine of complex numbers and the first beginnings of the theory of biquadratic residues; we shall undertake in a following continuation to make that theory complete on every side.¹

Art. 31

First of all we put forward certain names, whose introduction will serve brevity and clarity. The field of complex numbers $a + bi$ contains

I. real numbers, where $b = 0$, and among these, according to the nature of a ,

- 1) zero,
- 2) positive numbers,
- 3) negative numbers;

II. imaginary numbers, where b is unequal to zero. These are again divided into

¹It remains useful to observe here, at least in passing, that the field thus defined is especially adapted to the theory of biquadratic residues. The theory of cubic residues is similarly to be built upon the consideration of numbers of the form $a + bh$, where h is an imaginary root of the equation $h^3 - 1 = 0$, for instance $h = -\frac{1}{2} + \sqrt{\frac{3}{4}}i$; and likewise the theory of residues of higher powers will require the introduction of other imaginary quantities.

- 1) imaginary numbers without real part, that is, where $a = 0$,
- 2) imaginary numbers with real part, where neither b nor a is zero.

The former may, if desired, be called *pure imaginary numbers*; the latter, *mixed imaginary numbers*.

In this doctrine we use four units, $+1$, -1 , $+i$, $-i$, which will be called, simply, the positive, negative, positive imaginary, and negative imaginary units. The three products of any complex number by -1 , $+i$, $-i$ we shall call its *associates*, or the numbers *associated* with it. Except for zero, which is associated with itself, four unequal associated numbers therefore always occur.

Conversely, we call *conjugate* to a complex number the number that arises from it by replacing i by $-i$. Thus among imaginary numbers there are always two unequal conjugates, whereas real numbers are conjugate to themselves, if one chooses to extend the name to them.

The product of a complex number by its conjugate we call the *norm* of both. Thus the norm of a real number is to be taken as its square. In general we have eight connected numbers, namely

$$\begin{array}{c|c} a + bi & a - bi \\ -b + ai & -b - ai \\ -a - bi & -a + bi \\ b - ai & b + ai \end{array}$$

in which we see two quaternions of associated numbers and four pairs of conjugates; the common norm of all is $aa + bb$. But these eight numbers reduce to four unequal ones whenever either $a = \pm b$ or one of the numbers a , b is zero.

From these definitions the following facts follow immediately. The conjugate of a product of two complex numbers is the product of the numbers conjugate to them. The same is true of a product of several factors, and also of quotients. The norm of a product of two complex numbers is equal to the product of their norms. This theorem likewise extends to products of arbitrarily many factors and to quotients. The norm of any complex number, except zero, an exception that we shall henceforth usually understand tacitly, is a positive number.

There is, moreover, nothing to prevent our definitions from being extended to fractional or even irrational values of a and b ; but $a + bi$ will be called a *complex integral number* only when both a and b are integers, and it will be called *rational* only when both a and b are rational.

Art. 32

The algorithm of arithmetic operations on complex numbers is commonly known. Division is reduced to multiplication by the introduction of the norm, since

$$\frac{a + bi}{c + di} = (a + bi) \cdot \frac{c - di}{cc + dd} = \frac{ac + bd}{cc + dd} + \frac{bc - ad}{cc + dd} i.$$

Extraction of a square root is effected with the aid of the formula

$$\sqrt{a + bi} = \pm \left(\sqrt{\frac{\sqrt{aa + bb} + a}{2}} + i \sqrt{\frac{\sqrt{aa + bb} - a}{2}} \right)$$

if b is positive, or of the formula

$$\sqrt{a + bi} = \pm \left(\sqrt{\frac{\sqrt{aa + bb} + a}{2}} - i \sqrt{\frac{\sqrt{aa + bb} - a}{2}} \right)$$

if b is negative. There is no need here to dwell on the use of transforming the complex quantity $a + bi$ into $r(\cos \varphi + i \sin \varphi)$ in order to make calculations easier.

Art. 33

A complex integer that can be resolved into two factors different from units² we call a *composite complex number*; on the other hand, a number that admits no such resolution into factors will be called a *complex prime*. It follows at once that every composite real number is also composite as a complex number.

But a real prime number may be composite as a complex number. This is true of the number 2, and of all positive real primes of the form $4n + 1$ (apart from the number 1), since they are known to be decomposable into two positive squares; for example,

$$2 = (1 + i)(1 - i), \quad 5 = (1 + 2i)(1 - 2i), \quad 13 = (3 + 2i)(3 - 2i), \quad 17 = (1 + 4i)(1 - 4i), \quad \text{etc.}$$

On the other hand, positive real primes of the form $4n + 3$ are always complex primes. For if such a number q were equal to $(a + bi)(\alpha + \beta i)$, then also $q = (a - bi)(\alpha - \beta i)$, and hence

$$qq = (aa + bb)(\alpha\alpha + \beta\beta).$$

But qq can be resolved in only one way into positive factors greater than unity, namely as $q \times q$; hence one would have to have $q = aa + bb = \alpha\alpha + \beta\beta$, which is absurd, since a sum of two squares cannot be of the form $4n + 3$.

Negative real numbers plainly retain the same denominations as the corresponding positive ones, and the same is true of pure imaginary numbers.

It remains to teach how, among mixed imaginary numbers, composites are to be distinguished from primes. This is done by the following theorem.

THEOREM. Every mixed imaginary complex integer $a + bi$ is either a complex prime or a composite number, according as its norm is a real prime number or a composite number.

PROOF. I. Since the norm of a composite complex number is always composite, it is clear that a complex number whose norm is a real prime number must necessarily be a complex prime. This proves the first assertion.

II. If the norm $aa + bb$ is composite, let p be a positive real prime measuring it. Two cases must now be distinguished.

1) If p is of the form $4n + 3$, it is known that $aa + bb$ cannot be divisible by p unless p measures a and b themselves; hence $a + bi$ will be composite.

²Or, what is the same thing, factors whose norms are greater than unity.

2) If p is not of the form $4n + 3$, it can certainly be decomposed into two squares. We therefore put $p = \alpha\alpha + \beta\beta$. Since

$$(a\alpha + b\beta)(a\alpha - b\beta) = aa(\alpha\alpha + \beta\beta) - \beta\beta(aa + bb)$$

is therefore divisible by p , p must measure one of the two factors $a\alpha + b\beta$ and $a\alpha - b\beta$. Moreover, since

$$(a\alpha + b\beta)^2 + (b\alpha - a\beta)^2 = (a\alpha - b\beta)^2 + (b\alpha + a\beta)^2 = (aa + bb)(\alpha\alpha + \beta\beta)$$

is divisible by pp , it is clear that in the first case $b\alpha - a\beta$, and in the second case $b\alpha + a\beta$, must also be divisible by p . Therefore in the first case

$$\frac{a + bi}{\alpha + \beta i} = \frac{a\alpha + b\beta}{p} + \frac{b\alpha - a\beta}{p} i$$

will be a complex integer, while in the second case

$$\frac{a + bi}{\alpha - \beta i} = \frac{a\alpha - b\beta}{p} + \frac{b\alpha + a\beta}{p} i$$

will be an integer. Since the proposed number is thus divisible either by $\alpha + \beta i$ or by $\alpha - \beta i$, and since by hypothesis the norm of the quotient, $\frac{aa+bb}{p}$, is different from unity, it is clear in both cases that $a + bi$ is a composite complex number. This proves the second assertion.

Art. 34

The whole range of complex prime numbers is therefore exhausted by the following four species:

- 1) the four units $1, +i, -1, -i$, which, however, when we deal with prime numbers, we shall usually understand to be tacitly excluded;
- 2) the number $1 + i$, together with its three associates $-1 + i, -1 - i, 1 - i$;
- 3) positive real primes of the form $4n + 3$, together with their three associates;
- 4) complex numbers whose norms are real primes of the form $4n + 1$, greater than unity; moreover, to each such given norm there always correspond exactly eight complex primes, since such a norm can be decomposed into two squares in one way only.

Art. 35

Just as real integers are divided into even and odd numbers, and the even numbers again into evenly even and unevenly even, so among complex numbers an equally essential distinction presents itself. Namely, they are either

- not* divisible by $1 + i$, as are the numbers $a + bi$ where one of a, b is odd and the other even;
- or* divisible by $1 + i$ but not by 2, whenever both a and b are odd;
- or* divisible by 2, whenever both a and b are even.

Numbers of the first class may conveniently be called *odd complex numbers*; those of the second, *semieven*; those of the third, *even*.

A product of several complex factors will always be odd whenever all the factors are odd; semieven whenever one factor is semieven and the rest are odd; and even whenever among the factors there occur either at least two semieven factors or at least one even factor.

The norm of every odd complex number is of the form $4n + 1$; the norm of a semieven number is of the form $8n + 2$; finally, the norm of an even number is the product of a number of the form $4n + 1$ by 4 or by a higher power of two.

Art. 36

Since the connection among four associated complex numbers is analogous to the connection between two opposite real numbers, that is, numbers absolutely equal and affected with opposite signs, and since among such real numbers the positive one is commonly and rightly considered primary, the question arises whether a similar distinction can be established among four associated complex numbers, and whether it should be considered useful.

In order to decide this, one must observe that the principle of distinction ought to be so constituted that the product of two numbers that are primary among their associates is always primary among its associates. But we soon become certain that no such principle exists at all unless the distinction is restricted to integral numbers; indeed, the useful distinction must be limited to the odd numbers. For these, however, the proposed aim can be attained in two ways:

I. The product of two numbers $a + bi$ and $a' + b'i$ so arranged that a and a' are of the form $4n + 1$ and b and b' are even will enjoy the same property, so that the real part is $\equiv 1 \pmod{4}$ and the imaginary part is even. It is easy to see that among the four odd associates exactly one is contained under that form.

II. If the number $a + bi$ is so arranged that $a - 1$ and b are either both evenly even or both unevenly even, then its product by a complex number of the same form will enjoy the same form; and again it is easy to see that among the four associated odd numbers exactly one is contained under this form.

Of these two nearly equally suitable principles we shall adopt the latter: among the four associated odd complex numbers, we shall regard as primary that one which, modulo $2 + 2i$, becomes congruent to the positive unit. In this way several noteworthy theorems can be stated with greater neatness. Thus, for example, the primary complex primes are $-1 + 2i$, $-1 - 2i$, $+3 + 2i$, $+3 - 2i$, $+1 + 4i$, $+1 - 4i$, etc., and also the real primes -3 , -7 , -11 , -19 , etc., which must plainly always be taken with the negative sign. The conjugate of an odd primary complex number will also be primary.

For semieven and even numbers in general, a similar distinction would be too arbitrary and of little use. From the associated prime numbers $1 + i$, $1 - i$, $-1 + i$, $-1 - i$, we can indeed choose one as primary before the others; but we shall not extend such a distinction to composites.

Art. 37

If among the factors of a composite complex number there are some that are themselves composite, and if these are again resolved into their own factors, then one plainly comes at last to prime factors; that is, every composite number is resolvable into prime factors. If among these there are any that are not primary, substitute in place of each the product of the associated primary number by i , -1 , or $-i$. In this way it is clear that every composite complex number M can be reduced to the form

$$M = i^\mu A^\alpha B^\beta C^\gamma \dots,$$

where A, B, C , etc. are unequal primary complex primes, and $\mu = 0, 1, 2$, or 3 . Concerning this resolution the theorem presents itself that it can be made in one way only. Considered casually, this theorem might seem evident in itself, but it certainly needs a demonstration. The following prepares the way for it.

THEOREM. The product $M = A^\alpha B^\beta C^\gamma \dots$, where A, B, C , etc. denote distinct primary complex primes, cannot be divisible by any primary complex prime that is not found among A, B, C , etc.

PROOF. Let P be a primary complex prime not contained among A, B, C , etc., and let p, a, b, c , etc. be the norms of P, A, B, C , etc. It follows easily that the norm of M will be $a^\alpha b^\beta c^\gamma$, etc.; hence, if M were divisible by P , this number would have to be divisible by p .

Since the individual norms are either real prime numbers, from the series 2, 5, 13, 17, etc., or squares of real prime numbers, from the series 9, 49, 121, etc., it is immediately clear that this cannot occur unless p becomes identical with one of the norms a, b, c , etc. We therefore suppose $p = a$. But since P and A are, by hypothesis, non-identical primary complex primes, it is easily seen that these conditions can coexist only if P and A are conjugate imaginary complex numbers, and consequently $p = a$ is an odd real prime, not the square of a prime. We therefore suppose $A = k + li$ and $P = k - li$.

It follows, extending the notion and sign of congruence to complex integers, that $A \equiv 2k \pmod{P}$, whence one easily obtains

$$M \equiv 2^\alpha k^\alpha B^\beta C^\gamma \dots \pmod{P}.$$

Thus, while M is assumed to be divisible by P , the number $2^\alpha k^\alpha B^\beta C^\gamma \dots$ will also be divisible by P , and therefore the norm of this number, namely $2^{2\alpha} k^{2\alpha} b^\beta c^\gamma \dots$, will be divisible by p . But since neither 2 nor k is divisible by p , it follows that p must be identical with one of the numbers b, c , etc.; let it be, for example, $p = b$. From this, however, we conclude that either $B = k + li$ or $B = k - li$, that is, either $B = A$ or $B = P$, both contrary to the hypothesis.

From this theorem the other, namely that the resolution into prime factors can be carried out in one way only, is derived very easily, by arguments entirely analogous to those used for real numbers in the *Disquisitiones Arithmeticae*, Article 16. It would therefore be superfluous to dwell on them here.

Art. 38

We now proceed to the congruence of numbers with respect to complex moduli. But at the threshold of this investigation it is useful to indicate how the domain of complex quantities can be made subject to intuition.

Just as every real quantity can be expressed by a portion of a straight line, infinite in both directions, taken from an arbitrary origin and measured according to an arbitrary segment accepted as unit, and therefore can be represented by a point, with points on one side of the origin representing positive quantities and those on the other side negative quantities, so every complex quantity can be represented by some point in an infinite plane, in which a fixed straight line is referred to the real quantities. Specifically, the complex quantity $x + iy$ is represented by the point whose abscissa is x and whose ordinate, taken positively on one side of the line of abscissas and negatively on the other, is y .

In this way one may say that any complex quantity measures the difference between the position of the point to which it is referred and the position of the initial point; the positive unit denotes an arbitrary fixed displacement in an arbitrary fixed direction, the negative unit an equal displacement in the opposite direction, and finally the imaginary units equal displacements in the two normal lateral directions. In this way the metaphysics of the quantities that we call imaginary is strikingly illuminated.

If the initial point is denoted by (0) , and two complex quantities m and m' are referred to points M and M' , whose positions relative to (0) they express, then the difference $m - m'$ will be nothing other than the position of M relative to the point M' . Conversely, if the product mm' represents the position of the point N relative to (0) , one easily sees that this position is determined by the position of M with respect to (0) in the same way as the position of M' is determined by the position of the point corresponding to the positive unit; thus one may quite aptly say that the positions of the points corresponding to the complex quantities mm' , m , m' , and 1 form a proportion.

But we reserve a fuller treatment of this matter for another occasion. The difficulties in which the theory of the quantities we call imaginary is thought to be involved have for the most part arisen from ill-suited names, since some have even used the absurd name “impossible quantities.” If, taking our start from the concepts offered by varieties of two dimensions, which are seen in their greatest purity in the intuitions of space, we had called positive quantities direct, negative quantities inverse, and imaginary quantities lateral, then in place of tangles there would have been simplicity, and in place of darkness, clarity.

Art. 39

What was set forth in the preceding article refers to continuous complex quantities. In arithmetic, which is concerned only with integers, the diagram of complex numbers will be a system of equidistant points arranged on equidistant straight lines in such a way that the infinite plane is divided into infinitely many equal squares.

All numbers divisible by a given complex number $a + bi = m$ will likewise form infinitely many squares, whose sides are $= \sqrt{aa + bb}$ and whose areas are $= aa + bb$; these latter squares will be inclined to the former ones whenever neither a nor b is zero.

To every number not divisible by the modulus m there will correspond a point either situated within such a square or on a side common to two contiguous squares. The latter case, however, cannot occur unless a and b have a common divisor. Moreover, it is clear that numbers congruent modulo m occupy congruent positions in their squares.

From this it is easy to conclude that, if all numbers lying within a fixed square are collected, together with any that may lie on two of its non-opposite sides, and if to these one appends the number divisible by m , then one has a complete system of incongruent residues modulo m ; that is, every integer must be congruent to one and only one of them. It would not be difficult to show that the number of these residues is equal to the norm of the modulus, namely $aa + bb$. But it seems advisable to prove this very important theorem in another, purely arithmetic way.

Art. 40

THEOREM. With respect to a given complex modulus $m = a + bi$, whose norm is $aa + bb = p$, and for which a and b are relatively prime, every complex integer is congruent to one of the residues in the series $0, 1, 2, 3 \dots p - 1$, and to no more than one.

PROOF. I. Let α and β be integers such that $\alpha a + \beta b = 1$. Then

$$i = \alpha b - \beta a + m(\beta + \alpha i).$$

Thus for a proposed complex integer $A + Bi$ we have

$$A + Bi = A + (\alpha b - \beta a)B + m(\beta B + \alpha Bi).$$

Therefore, denoting by h the least nonnegative residue of the number $A + (\alpha b - \beta a)B$ modulo p , and putting

$$A + (\alpha b - \beta a)B = h + kp = h + m(ak - bki),$$

we obtain

$$A + Bi = h + m(\beta B + ak + (\alpha B - bk)i),$$

or $A + Bi \equiv h \pmod{m}$. This proves the first assertion.

II. Whenever two real numbers h and h' are congruent to the same complex number modulo m , they are also congruent to one another. Put $h - h' = m(c + di)$; then

$$(h - h')(a - bi) = p(c + di),$$

and hence

$$(h - h')a = pc, \quad (h - h')b = -pd.$$

Moreover, since $a\alpha + b\beta = 1$,

$$h - h' = p(c\alpha - d\beta), \quad \text{that is,} \quad h \equiv h' \pmod{p}.$$

Consequently, if h and h' are unequal, they cannot both be contained in the set of numbers $0, 1, 2, 3 \dots p - 1$. This proves the second assertion.

Art. 41

THEOREM. With respect to the complex modulus $m = a + bi$, whose norm is $aa + bb = p$, and for which a and b are not relatively prime but have greatest common divisor λ , taken positive, every complex number is congruent to a residue $x + yi$ such that x is one of the numbers

$$0, 1, 2, 3 \dots \frac{p}{\lambda} - 1,$$

and y is one of $0, 1, 2, 3 \dots \lambda - 1$; moreover, it is congruent to exactly one among all the p residues that have this form.

PROOF. I. Choose integers α, β such that $\alpha a + \beta b = \lambda$. Then

$$\lambda i = \alpha b - \beta a + m(\beta + \alpha i).$$

Now let $A + Bi$ be the proposed complex number, let y be the least nonnegative residue of B modulo λ , and let x be the least nonnegative residue of

$$A + (\alpha b - \beta a) \cdot \frac{B - y}{\lambda}$$

modulo $\frac{p}{\lambda}$; and put

$$A + (\alpha b - \beta a) \cdot \frac{B - y}{\lambda} = x + \frac{p}{\lambda} \cdot k.$$

It follows that

$$\begin{aligned} A + Bi - (x + yi) &= \frac{p}{\lambda} \cdot k + (B - y)i - (\alpha b - \beta a) \frac{B - y}{\lambda} \\ &= \frac{p}{\lambda} \cdot k + \frac{B - y}{\lambda} \cdot m(\beta + \alpha i) \\ &= \left(\frac{a}{\lambda} - \frac{b}{\lambda} i \right) km + \frac{B - y}{\lambda} (\beta + \alpha i)m, \end{aligned}$$

that is, the difference is divisible by m , or $A + Bi \equiv x + yi \pmod{m}$. This proves the first assertion.

II. Suppose that two numbers $x + yi$ and $x' + y'i$ are congruent modulo m to the same complex number. Then they are also congruent to one another modulo m . A fortiori they are congruent modulo λ , and therefore $y \equiv y' \pmod{\lambda}$. If each of y and y' is supposed to be contained among the numbers $0, 1, 2, 3 \dots \lambda - 1$, one must have $y = y'$.

But then also $x \equiv x' \pmod{m}$; that is, $x - x'$ is divisible by m , and consequently the integer $\frac{x - x'}{\lambda}$ is divisible by $\frac{a}{\lambda} + \frac{b}{\lambda} i$, or

$$\frac{x - x'}{\lambda} \equiv 0 \pmod{\frac{a}{\lambda} + \frac{b}{\lambda} i}.$$

Since $\frac{a}{\lambda}$ and $\frac{b}{\lambda}$ are relatively prime, it follows by the second part of the theorem of the preceding article that $\frac{x - x'}{\lambda}$ is also divisible by the norm of $\frac{a}{\lambda} + \frac{b}{\lambda} i$, that is, by $\frac{p}{\lambda\lambda}$, and hence that $x - x'$ is divisible by $\frac{p}{\lambda}$. Therefore, if each of x and x' is also supposed to be contained in the set of numbers $0, 1, 2, 3 \dots \frac{p}{\lambda} - 1$, it is necessary that $x = x'$; that is, the residues $x + yi$ and $x' + y'i$ are identical. This proves the second assertion.

It is, moreover, clear at once that here too one must include the case in which the modulus is a real number, say $b = 0$ and hence $\lambda = \pm a$, as well as the case in which the modulus is purely imaginary, say $a = 0$ and hence $\lambda = \pm b$. In both cases one has $\frac{p}{\lambda} = \lambda$.

Art. 42

Thus, if all complex integers that are congruent to one another with respect to a given modulus are referred to the same class, and incongruent integers to different classes, there will be altogether p classes exhausting the whole range of integers, where p denotes the norm of the modulus. A collection of just as many numbers, taken one from each class, will give a complete system of incongruent residues, as we assigned in Articles 40 and 41.

In that system the choice of the residues that represent their classes was based on this principle: in each class one adopted a residue $x + yi$ for which y had the least value; and, among all residues having this same least value of y , one chose the one for which the value of x was least, negative values being excluded both for x and for y .

For other purposes it will be better to use other principles. Especially to be noted is the method in which those residues are adopted which, when divided by the modulus, give the simplest quotients. Plainly, if

$$\alpha + \beta i, \quad \alpha' + \beta' i, \quad \alpha'' + \beta'' i, \dots$$

are quotients arising from division of congruent numbers by the modulus, then the differences among the quantities $\alpha, \alpha', \alpha'', \dots$ are integers, and so also are the differences among $\beta, \beta', \beta'', \dots$. Hence there is always a single residue for which α and β lie between 0 and 1, the former limit included and the latter excluded. Such a residue we call simply a *minimum residue*.

If preferred, instead of those limits one may adopt $-\frac{1}{2}$ and $+\frac{1}{2}$, admitting one and excluding the other. The residue corresponding to this restriction we shall call *absolutely minimum*. The following problems present themselves concerning these minimum residues.

Art. 43

The minimum residue of a given complex number $A + Bi$ with respect to the modulus $a + bi$, whose norm is p , is found as follows. If $x + yi$ is the required minimum residue, then $(x + yi)(a - bi)$ will be the minimum residue of the product $(A + Bi)(a - bi)$ with respect to the modulus $(a + bi)(a - bi)$, that is, with respect to the modulus p .

Therefore put

$$aA + bB = Fp + f, \quad aB - bA = Gp + g,$$

so that f and g are minimum residues of $aA + bB$ and $aB - bA$ modulo p . Then

$$x + yi = \frac{f + gi}{a - bi},$$

or

$$x = \frac{af - bg}{p} = A - aF + bG, \quad y = \frac{ag + bf}{p} = B - aG - bF.$$

Obviously the minimum residues f, g must be taken either between 0 and $p - 1$, or between $-\frac{1}{2}p$ and $+\frac{1}{2}p$, according as a simply minimum or an absolutely minimum complex residue is desired.

Art. 44

A complete system of minimum residues for a given modulus can be constructed in several ways. The first method proceeds by first determining the limits within which the real terms must lie, and then, for each value lying within those limits, assigning the limits for the imaginary parts.

The general criterion for a minimum residue $x + yi$ with respect to the modulus $a + bi$ consists in this: both $ax + by = \xi$ and $ay - bx = \eta$ must lie between the limits 0 and $aa + bb$ when simply minimum residues are in question, or between the limits $-\frac{1}{2}(aa + bb)$ and $+\frac{1}{2}(aa + bb)$ when absolutely minimum residues are desired, one of the two endpoints being excluded.

Special rules would require a distinction of the cases introduced by the various possible signs of a and b . Since that development presents no difficulty, we shall not dwell on it here; it will be enough to display the nature of the method by a single example.

For the modulus $5 + 2i$, the simply minimum residues $x + yi$ must be so chosen that both $5x + 2y = \xi$ and $5y - 2x = \eta$ are equal to one of the numbers $0, 1, 2, 3, \dots, 28$.

The equation $29x = 5\xi - 2\eta$ shows that positive values of x cannot be greater than $\frac{5 \cdot 28}{29}$, and that negative values, disregarding the sign, cannot be greater than $\frac{2 \cdot 28}{29}$. Thus all admissible values of x are $-1, 0, 1, 2, 3, 4$.

For $x = -1$, $2y$ must be equal to one of the numbers $5, 6, 7, \dots, 33$, while $5y$ must be equal to one of the numbers $-2, -1, 0, 1, \dots, 26$; hence the least value of y is $+3$, and the greatest is $+5$.

Treating the remaining values of x in the same way, one obtains the following scheme of all minimum residues:

x	y
-1	$3, 4, 5$
0	$0, 1, 2, 3, 4, 5$
$+1$	$1, 2, 3, 4, 5, 6$
$+2$	$1, 2, 3, 4, 5, 6$
$+3$	$2, 3, 4, 5, 6$
$+4$	$2, 3, 4$

Similarly, for absolutely minimum residues, ξ and η must be equal to one of the numbers $-14, -13, -12, \dots, +14$. Hence $29x$ cannot fall outside the limits $-7 \cdot 14$ and $+7 \cdot 14$, and therefore x must be equal to one of the numbers $-3, -2, -1, 0, 1, 2, 3$.

For $x = -3$, $2y = \xi - 5x = \xi + 15$ will be equal to one of the numbers $1, 2, 3, \dots, 29$, while $5y = \eta + 2x = \eta - 6$ will be equal to one of the numbers $-20, -19, -18, \dots, +8$. Hence there results for y the single value $+1$.

Treating the remaining values of x in the same way, we obtain the scheme of all absolutely

minimum residues:

x	y
-3	+1
-2	-2, -1, 0, +1, +2
-1	-3, -2, -1, 0, +1, +2
0	-2, -1, 0, +1, +2
+1	-2, -1, 0, +1, +2, +3
+2	-2, -1, 0, +1, +2
+3	-1

Art. 45

In applying the second method it will be convenient to distinguish two cases.

In the first case, where a and b have no common divisor, put

$$\alpha a + \beta b = 1,$$

and let k be the least positive residue of $\beta a - \alpha b$ modulo p . From the identities

$$a(\beta a - \alpha b) = \beta p - b(\alpha a + \beta b), \quad b(\beta a - \alpha b) = -\alpha p + a(\alpha a + \beta b)$$

it follows that

$$ak \equiv -b, \quad bk \equiv a \pmod{p}.$$

Putting, as above,

$$ax + by = \xi, \quad ay - bx = \eta,$$

we therefore have

$$\eta \equiv k\xi, \quad \xi \equiv -k\eta \pmod{p}.$$

Accordingly, all numbers $\xi + \eta i$ to which simply minimum residues $x + yi$ correspond will be obtained either by taking for ξ successively the values $0, 1, 2, 3, \dots, p-1$ and for η the least positive residues of the products $k\xi$ modulo p , or, in the other order, by taking those values for η and for ξ the least residues of the products $-k\eta$.

From each $\xi + \eta i$ the corresponding $x + yi$ is then found by the formula

$$x + yi = \frac{\xi + \eta i}{a - bi} = \frac{a\xi - b\eta}{p} + \frac{a\eta + b\xi}{p}i.$$

Moreover, it is clear that when ξ is increased by one, η undergoes either an increase by k or a decrease by $p - k$, and therefore $x + yi$ undergoes either the change

$$\frac{a - kb}{p} + \frac{ak + b}{p}i$$

or this one:

$$\frac{a - kb}{p} + b + \left(\frac{ak + b}{p} - a \right) i.$$

This observation is useful for making the construction easier.

Finally, if absolutely minimum residues $x + yi$ are wanted, the preceding rules need be changed only to this extent: the values successively assigned to ξ should lie between $-\frac{1}{2}p$ and $+\frac{1}{2}p$, while η should be taken as the absolutely minimum residue of the products $k\xi$.

Here is the display of the minimum residues for the modulus $5 + 2i$ arranged by this method.

Simply minimum residues.

$\xi + \eta i$	$x + yi$	$\xi + \eta i$	$x + yi$	$\xi + \eta i$	$x + yi$
0	0	10 + 25i	+5i	20 + 21i	+2 + 5i
1 + 17i	-1 + 3i	11 + 13i	+1 + 3i	21 + 9i	+3 + 3i
2 + 5i	+i	12 + i	+2 + i	22 + 26i	+2 + 6i
3 + 22i	+1 + 4i	13 + 18i	+1 + 4i	23 + 14i	+3 + 4i
4 + 10i	+2i	14 + 6i	+2 + 2i	24 + 2i	+4 + 2i
5 + 27i	-1 + 5i	15 + 23i	+1 + 5i	25 + 19i	+3 + 5i
6 + 15i	+3i	16 + 11i	+2 + 3i	26 + 7i	+4 + 3i
7 + 3i	+1 + i	17 + 28i	+1 + 6i	27 + 24i	+3 + 6i
8 + 20i	+4i	18 + 16i	+2 + 4i	28 + 12i	+4 + 4i
9 + 8i	+1 + 2i	19 + 4i	+3 + 2i		

Absolutely minimum residues.

$\xi + \eta i$	$x + yi$	$\xi + \eta i$	$x + yi$	$\xi + \eta i$	$x + yi$
-14 - 6i	-2 - 2i	-4 - 10i	-2i	+5 - 2i	+1
-13 + 11i	-3 + i	-3 + 7i	-1 + i	+6 - 14i	+2 - 2i
-12 - i	-2 - i	-2 - 5	-i	+7 + 3i	+1 + i
-11 - 13i	-1 - 3i	-1 + 12i	-1 + 2i	+8 - 9i	+2 - i
-10 + 4i	-2	0	0	+9 + 8i	+1 + 2i
-9 - 8i	-1 - 2i	+1 - 12i	+1 - 2i	+10 - 4i	+2
-8 + 9i	-2 + i	+2 + 5i	+i	+11 + 13i	+1 + 3i
-7 - 3i	-1 - i	+3 - 7i	+1 - i	+12 + i	+2 + i
-6 + 14i	-2 + 2i	+4 + 10i	+2i	+13 - 11i	+3 - i
-5 + 2i	-1			+14 + 6i	+2 + 2i

The second case, where a, b are not relatively prime, is easily reduced to the preceding case. Let λ be the greatest common divisor of a, b , and let $a = \lambda a', b = \lambda b'$. Let F denote, indefinitely, a minimum residue for the modulus λ , considered as a complex number; that is, let it denote a number $x + yi$ such that x, y lie either between 0 and λ , or between $-\frac{1}{2}\lambda$ and $+\frac{1}{2}\lambda$, according as one is dealing with simply or absolutely minimum residues. Let F' likewise denote, indefinitely, a minimum residue for the modulus $a' + b'i$. Then

$$(a' + b'i)F + F'$$

will be, indefinitely, a minimum residue for the modulus $a + bi$; and a complete system of such residues will be produced by combining every F with every F' .

Art. 46

Two complex numbers are said to be relatively prime if, apart from the units, they admit no other common divisors. Whenever such common divisors are present, those common divisors whose norm is greatest are called greatest common divisors.

If the decomposition of the two proposed numbers into prime factors is at hand, the determination of the greatest common divisor is carried out exactly as for real integers (*Disquisitiones Arithmeticae*, Art. 18). At the same time it follows that every common divisor of the two given numbers must divide the greatest common divisor found in this way.

Since it is already obvious that the three associates of such a number are also common divisors, exactly four numbers, and no more, will always be called greatest common divisors; their norm will be a multiple of the norm of any other common divisor.

If the decomposition of the two proposed numbers into simple factors is not available, the greatest common divisor is extracted by means of an algorithm similar to the one used for real integers. Let m, m' be the two proposed numbers, and form by repeated division the sequence $m'', m''', \dots$, so that m'' is the absolutely minimum residue of m modulo m' , then m''' the absolutely minimum residue of m' modulo m'' , and so on.

Denoting the norms of the numbers $m, m', m'', m''', \dots$ by $p, p', p'', p''', \dots$ respectively, p''/p' is the norm of the quotient m''/m' and therefore, by the definition of the absolutely minimum residue, is certainly not greater than $\frac{1}{2}$. The same holds for p'''/p'' , and so forth. Consequently the positive real integers $p', p'', p''', \dots$ form a continually decreasing sequence, and therefore one must finally reach the term 0.

Let the preceding final divisor be $m^{(n+1)}$, and put

$$\begin{aligned} m &= km' + m'', \\ m' &= k'm'' + m''', \\ m'' &= k''m''' + m''', \end{aligned}$$

and so on, up to

$$m^{(n)} = k^{(n)}m^{(n+1)}.$$

Running through these equations in the reverse order, it is clear that $m^{(n+1)}$ divides all preceding terms $m^{(n)}, \dots, m'', m', m$; running through them in the direct order, it is clear that any common divisor of m, m' also divides all following terms. The first conclusion says that $m^{(n+1)}$ is a common divisor of m, m' ; the second says that this divisor is greatest.

Finally, whenever the last residue $m^{(n+1)}$ turns out to be equal to one of the four units $1, -1, i, -i$, this is a sign that m and m' are relatively prime.

Art. 47

If the equations of the preceding article, with the last omitted, are combined so that $m'', m''', m''', \dots, m^{(n)}$ are eliminated, an equation of the form

$$m^{(n+1)} = hm + h'm'$$

will arise, where h, h' are integers. In the notation introduced in Disq. Arith. Art. 27,

$$\begin{aligned} h &= \pm[k', k'', k''', \dots, k^{(n-1)}] = \pm[k^{(n-1)}, k^{(n-2)}, \dots, k'', k'], \\ h' &= \mp[k, k', k'', k''', \dots, k^{(n-1)}] = \mp[k^{(n-1)}, k^{(n-2)}, \dots, k'', k', k], \end{aligned}$$

the upper or lower signs holding according as n is even or odd.

We state this theorem as follows: the greatest common divisor of two complex numbers m, m' can be reduced to the form $hm + h'm'$, where h, h' are integers.

For this holds not only for the greatest common divisor to which the algorithm of the preceding article has led, but also for its three associates. For those, in place of the coefficients h, h' one must take either $hi, h'i$, or $-h, -h'$, or $-hi, -h'i$.

Therefore whenever m, m' are relatively prime, it will be possible to satisfy the equation

$$1 = hm + h'm'.$$

For example, suppose the proposed numbers are $31 + 6i = m$ and $11 - 20i = m'$. We find

$$\begin{aligned} k &= i, & m'' &= +11 - 5i, \\ k' &= +1 - i, & m''' &= +5 - 4i, \\ k'' &= +2, & m'''' &= +1 + 3i, \\ k''' &= -1 - 2i, & m''''' &= +i, \\ k'''' &= +3 - i. \end{aligned}$$

Hence

$$[k', k'', k'''] = -6 - 5i, \quad [k, k', k'', k'''] = +4 - 10i,$$

and therefore

$$m''''' = i = (6 + 5i)m + (4 - 10i)m',$$

as well as

$$1 = (5 - 6i)m + (-10 - 4i)m',$$

which is confirmed by carrying out the calculation.

Art. 48

All that is required for the theory of first-degree congruences in the arithmetic of complex integers has now been prepared. Since that theory does not essentially differ from the corresponding theory for real integers, and since it has been fully explained in the *Disquisitiones Arithmeticae*, it will be enough to record the principal points here.

I. The congruence

$$mt \equiv 1 \pmod{m'}$$

is equivalent to the indeterminate equation

$$mt + m'u = 1.$$

If this equation is satisfied by $t = h, u = h'$, the solution of the congruence is expressed generally by

$$t \equiv h \pmod{m'}.$$

The condition of solvability is that the modulus m' have no common divisor with the coefficient m .

II. The solution of the congruence

$$ax + b \equiv c \pmod{M}$$

in the case where a, M are relatively prime depends on the solution of

$$at \equiv 1 \pmod{M}.$$

If this is satisfied by $t = h$, the general solution of the proposed congruence is contained in

$$x \equiv (c - b)h \pmod{M}.$$

III. The congruence $ax + b \equiv c \pmod{M}$ in the case where a, M have the common divisor λ is equivalent to

$$\frac{a}{\lambda}x \equiv \frac{c - b}{\lambda} \pmod{M/\lambda}.$$

Therefore, when λ is taken to be the greatest common divisor of a, M , the solution of the proposed congruence is reduced to the preceding case; and it is clear that, for solvability, it is necessary and sufficient that λ also divide the difference $c - b$.

Art. 49

Up to this point we have touched only elementary matters, which nevertheless could not be omitted because of the connection. In higher investigations the arithmetic of complex integers is like the arithmetic of real integers in this respect: more elegant and simpler theorems appear when only such moduli as are prime numbers are admitted. In truth, the extension of these theorems to composite moduli is usually longer rather than more difficult, more a matter of labor than of art. Therefore, in what follows, prime moduli will chiefly be considered.

Art. 50

Let X denote a function of the indeterminate x of the form

$$Ax^n + Bx^{n-1} + Cx^{n-2} + \text{etc.} + Mx + N,$$

where n is a positive real integer, $A, B, C, \dots$ are real or imaginary integers, and m is a complex integer. We shall here also call a root of the congruence

$$X \equiv 0 \pmod{m}$$

any integer which, when substituted for x , gives to X a value divisible by the modulus m . Solutions by roots congruent modulo the modulus will not be regarded as different.

Whenever the modulus is prime, such a congruence of order n can here too admit no more than n different solutions.

Let α be any fixed complex integer. By division by $x - \alpha$, the function X can be reduced indefinitely to the form

$$X = (x - \alpha)X' + h,$$

where h is a fixed integer and X' is a function of order $n - 1$ with integral coefficients. Whenever α is a root of the congruence $X \equiv 0 \pmod{m}$, h will plainly be divisible by m , or

$$X \equiv (x - \alpha)X' \pmod{m}.$$

Similarly, if for a fixed integer β the function X' is reduced to $(x - \beta)X'' + h'$, then X'' is of order $n - 2$ with integral coefficients. If β is also assumed to be a root of the congruence $X \equiv 0$, it must satisfy $(\beta - \alpha)X' \equiv 0$, and also $X' \equiv 0$, provided the roots α, β are incongruent. Therefore h' too must be divisible by m , or

$$X \equiv (x - \alpha)(x - \beta)X'' \pmod{m}.$$

In the same way, with a third root γ incongruent to the preceding ones, we obtain indefinitely

$$X \equiv (x - \alpha)(x - \beta)(x - \gamma)X''',$$

where X''' has order $n - 3$ and integral coefficients. Continuing in the same manner, and noting that the coefficient of the highest term in each function is A - which may be assumed not divisible by m , since otherwise the congruence $X \equiv 0$ would essentially belong to a lower order - we get, whenever there are n incongruent roots $\alpha, \beta, \gamma, \dots, \nu$,

$$X \equiv A(x - \alpha)(x - \beta)(x - \gamma) \cdots (x - \nu) \pmod{m}.$$

Thus substitution of a new value incongruent to all of $\alpha, \beta, \gamma, \dots, \nu$ would certainly give X a value not divisible by m . The theorem follows at once. This proof agrees essentially with that given in Disq. Arith. Art. 43, and each of its steps is valid for complex integers just as for real ones.

Art. 51

What was set forth in the third section of the Disquisitiones Arithmeticae concerning power residues remains valid, with only slight changes, for the most part also in the arithmetic of complex integers. Indeed, the demonstrations of the theorems could usually be retained. So that nothing be lacking, however, we shall give the principal theorems supported by concise proofs, always understanding that the modulus is a prime number.

Theorem. Let k be an integer not divisible by the modulus m , whose norm is p . Then

$$k^{p-1} \equiv 1 \pmod{m}.$$

Proof. Let $a, b, c, \dots$ constitute a complete system of incongruent residues modulo m , with the residue divisible by m omitted. Thus the number of these residues, which we denote collectively by C , is $p - 1$. Let C' be the collection of products $ka, kb, kc, \dots$

None of these products is divisible by m by hypothesis. Therefore each will have a congruent residue in C , say

$$ak \equiv a', \quad bk \equiv b', \quad ck \equiv c', \dots \pmod{m},$$

where $a', b', c', \dots$ are themselves found in C . Denote this collection by C'' .

Let P, P', P'' be the products of the numbers in the collections C, C', C'' respectively, that is,

$$P = abc \dots, \quad P' = k^{p-1}abc \dots = k^{p-1}P, \quad P'' = a'b'c' \dots.$$

Since the numbers of C'' are successively congruent to the numbers of C' , we have $P'' \equiv P'$, or $P'' \equiv k^{p-1}P$. But it is easy to see that any two numbers in C'' are incongruent, and hence all distinct; necessarily the collection C'' agrees exactly with C , only in changed order. Thus $P'' = P$.

Consequently $(k^{p-1} - 1)P$ is divisible by m . Since m is prime and does not divide any single factor of P , it necessarily follows that $k^{p-1} - 1$ must be divisible by m . Q.E.D.

Art. 52

Theorem. Let k , as in the preceding article, be an integer not divisible by the modulus m , and let t be the least positive exponent for which

$$k^t \equiv 1 \pmod{m}.$$

Then t divides any other exponent u for which $k^u \equiv 1 \pmod{m}$.

Proof. If t did not divide u , let gt be the multiple of t immediately greater than u ; then $gt - u$ is a positive integer smaller than t . From $k^t \equiv 1$ and $k^u \equiv 1$ it follows that

$$0 \equiv k^{gt} - k^u \equiv k^u(k^{gt-u} - 1),$$

so that $k^{gt-u} \equiv 1$. This gives a power of k congruent to unity with exponent smaller than t , contrary to the hypothesis.

As a corollary, t certainly divides $p - 1$. Those numbers k for which $t = p - 1$ we shall here also call primitive roots modulo m ; that such numbers actually exist we now show.

Art. 53

Resolve the number $p - 1$ into its prime factors, so that

$$p - 1 = a^\alpha b^\beta c^\gamma \dots,$$

where $a, b, c, \dots$ denote distinct positive real primes. Let $A, B, C, \dots$ be complex integers not divisible by m , and respectively not satisfying the congruences

$$x^{(p-1)/a} \equiv 1, \quad x^{(p-1)/b} \equiv 1, \quad x^{(p-1)/c} \equiv 1, \quad \text{etc.}$$

modulo m ; that such choices exist is clear from the theorem of Article 50. Finally let h be congruent modulo m to the product

$$A^{(p-1)/a^\alpha} B^{(p-1)/b^\beta} C^{(p-1)/c^\gamma} \dots.$$

I say that h will be a primitive root.

Proof. Denote by t the exponent of the least power h^t congruent to unity. If h were not a primitive root, t would be a proper divisor of $p-1$, so $(p-1)/t$ would be an integer greater than one. Plainly this integer has among its prime factors one of $a, b, c, \dots$. We may therefore suppose that $(p-1)/t$ is divisible by a , and put $p-1 = atu$.

Since $h^t \equiv 1$, also $h^{tu} \equiv 1$, that is,

$$A^{\frac{p-1}{a^\alpha} \cdot \frac{p-1}{a}} B^{\frac{p-1}{b^\beta} \cdot \frac{p-1}{a}} C^{\frac{p-1}{c^\gamma} \cdot \frac{p-1}{a}} \dots \equiv 1.$$

But $(p-1)/(ab^\beta)$ is plainly an integer, and hence

$$B^{\frac{p-1}{b^\beta} \cdot \frac{p-1}{a}} = (B^{p-1})^{(p-1)/(ab^\beta)} \equiv 1.$$

The same is true of the remaining factors except the power of A . Therefore we must have

$$A^{\frac{p-1}{a^\alpha} \cdot \frac{p-1}{a}} \equiv 1.$$

Now choose a positive integer λ such that

$$\lambda b^\beta c^\gamma \dots \equiv 1 \pmod{a},$$

which can be done because the prime a does not divide $b^\beta c^\gamma \dots$; write

$$\lambda b^\beta c^\gamma \dots = 1 + a\mu.$$

Then

$$A^{\lambda \cdot \frac{p-1}{a^\alpha} \cdot \frac{p-1}{a}} \equiv 1.$$

But

$$\lambda \cdot \frac{p-1}{a^\alpha} \cdot \frac{p-1}{a} = (1 + a\mu) \frac{p-1}{a} = (p-1)\mu + \frac{p-1}{a}.$$

Thus

$$A^{(p-1)\mu} A^{(p-1)/a} \equiv 1.$$

Since $A^{(p-1)\mu} \equiv 1$, it follows that

$$A^{(p-1)/a} \equiv 1,$$

contrary to the hypothesis. Therefore the supposition that t is a proper divisor of $p-1$ cannot stand; necessarily h is a primitive root.

Continuation: Articles 54–65

Art. 54

Let h denote a primitive root modulo m , whose norm is p . The terms of the progression

$$1, h, hh, h^3, \dots, h^{p-2}$$

will be mutually incongruent. Hence it is easily inferred that every integer not divisible by the modulus must be congruent to one of these; in other words, this series gives a complete system of incongruent residues, zero excluded. The exponent of the power to which a given number is congruent may be called its *index*, while h is regarded as the *base*. Here are some examples, in which the absolutely least residue has been put next to each index.

First example.

$$m = 5 + 4i, \quad p = 41, \quad h = 1 + 2i$$

Ind.	Residuuum	Ind.	Residuuum	Ind.	Residuuum	Ind.	Residuuum	Ind.	Residuuum
0	+1	8	-4	16	-2 + 2i	24	+2i	32	+1 + i
1	+1 + 2i	9	-3 + i	17	-1 + 2i	25	-3i	33	+1 + 3i
2	+1 - i	10	-i	18	+4i	26	+2 + 2i	34	+2
3	+3 + i	11	+2 - i	19	+1 + 3i	27	+2 + i	35	-3
4	-2i	12	-1 - i	20	-1	28	+4	36	+2 - 2i
5	+3i	13	+1 - 3i	21	-1 - 2i	29	+3 - i	37	+1 - 2i
6	-2 - 2i	14	-2	22	-1 + i	30	+i	38	-4i
7	-2 - i	15	+3	23	-3 - i	31	-2 + i	39	-1 - 3i

Second example.

$$m = 7, \quad p = 49, \quad h = 1 + 2i$$

Ind.	Residuuum	Ind.	Residuuum	Ind.	Residuuum	Ind.	Residuuum	Ind.	Residuuum
0	+1	10	-1 - i	20	+2i	30	+2 - 2i	40	+3
1	+1 + 2i	11	+1 - 3i	21	+3 + 2i	31	-1 + 2i	41	+3 - i
2	-3 - 3i	12	-i	22	-1 + i	32	+2	42	-2 - 2i
3	+3 - 2i	13	+2 - i	23	-3 - i	33	+2 - 3i	43	+2 + i
4	-3i	14	-3 + 3i	24	-1	34	+1 + i	44	-2i
5	-1 - 3i	15	-2 - 3i	25	-1 - 2i	35	-1 + 3i	45	-3 - 2i
6	-2 + 2i	16	-3	26	+3 + 3i	36	+i	46	+1 - i
7	+1 - 2i	17	-3 + i	27	-3 + 2i	37	-2 + i	47	+3 + i
8	-2	18	+2 + 2i	28	+3i	38	+3 - 3i		
9	-2 + 3i	19	-2 - i	29	+1 + 3i	39	+2 + 3i		

Art. 55

We add a few observations about primitive roots and the algorithm of indices, omitting the proofs because of their ease.

- I. Indices congruent modulo $p - 1$ correspond, in the given system, to residues congruent modulo m , and conversely.
- II. The residues corresponding to indices prime to $p - 1$ are themselves primitive roots, and conversely.
- III. If, with the primitive root h taken as base, the index of another primitive root h' is t , and conversely t' is the index of h when h' is taken as base, then $tt' \equiv 1 \pmod{p - 1}$; and, under the same assumptions, if the indices of some other number in these two systems are respectively u and u' , then $tu' \equiv u$ and $t'u \equiv u' \pmod{p - 1}$.
- IV. When the numbers $1, 1 + i$, and their three associates are excluded from the moduli under consideration, as being too meagre, there remain those primes that we placed third and fourth in Article 34. The norms of the latter will be real primes of the form $4n + 1$; the norms of the former are squares of odd real primes. Thus in both cases $p - 1$ is divisible by 4.
- V. If the index of the number -1 is denoted by u , then $2u \equiv 0 \pmod{p - 1}$, and hence either $u \equiv 0$ or $u \equiv \frac{1}{2}(p - 1)$. But since index 0 corresponds to the residue $+1$, the index of -1 must necessarily be $\frac{1}{2}(p - 1)$.
- VI. Similarly, if u denotes the index of i , then $2u \equiv \frac{1}{2}(p - 1) \pmod{p - 1}$, and hence either $u \equiv \frac{1}{4}(p - 1)$ or $u \equiv \frac{3}{4}(p - 1)$. This ambiguity depends on the choice of primitive root. Namely, if, with primitive root h taken as base, the index of i is $\frac{1}{4}(p - 1)$, the index becomes $\frac{3}{4}(p - 1)$ when h^μ is taken as base, where μ is a positive integer of the form $4n + 3$ and prime to $p - 1$, for instance the number $p - 2$ itself, and conversely. Thus one half of the primitive roots gives i the index $\frac{1}{4}(p - 1)$, the other half the index $\frac{3}{4}(p - 1)$; plainly, for the former bases $-i$ will have index $\frac{3}{4}(p - 1)$, and for the latter it will have index $\frac{1}{4}(p - 1)$.
- VII. Whenever the modulus is a positive real prime of the form $4n + 3$, say q , so that $p = qq$, the indices of all real numbers are divisible by $q + 1$. For if t denotes the index of the real number k , then from $k^{q-1} \equiv 1 \pmod{q}$ we get $(q - 1)t \equiv 0 \pmod{qq - 1}$, and hence $t/(q + 1)$ is an integer. Similarly the indices of purely imaginary numbers such as ki will be divisible by $\frac{1}{2}(q + 1)$. Thus primitive roots for such moduli must be sought only among mixed numbers.
- VIII. By contrast, for a modulus m that is a mixed complex prime, whose norm p is therefore a real prime of the form $4n + 1$, primitive roots can even be chosen among the real numbers, among which a complete system of incongruent residues may be displayed (Article 40). Plainly every real number that is a primitive root modulo the complex modulus m is at the same time, in the arithmetic of real numbers, a primitive root modulo p , and conversely.

Art. 56

Although the theory of quadratic residues and non-residues in the arithmetic of complex numbers is contained in the very theory of biquadratic residues, nevertheless, before passing to the latter, we shall present separately here the principal theorems of the former. For brevity we shall speak here only of the principal case, in which the modulus is an odd complex prime.

Let m be such a modulus, and let p be its norm. Plainly every integer not divisible by m - which will always be understood here - either can or cannot be made congruent to a square modulo m , according as its index, with some primitive root taken as base, is even or odd. In the former case that integer will be called a quadratic residue of m ; in the latter, a non-residue. It follows that among the $p - 1$ numbers forming a complete system of incongruent residues not divisible by m , one half are quadratic residues and the other half quadratic non-residues. Every number outside that system is to be given, in this respect, the same character as the number of the system to which it is congruent.

It also follows from the same source that the product of two quadratic residues, as well as the product of two non-residues, is a quadratic residue; whereas the product of a quadratic residue and a non-residue is a non-residue. More generally, a product of any number of factors is a quadratic residue or a non-residue according as the number of non-residue factors is even or odd.

For distinguishing quadratic residues from non-residues, the following general criterion at once presents itself:

$$k \text{ is a quadratic residue or non-residue} \quad \Longleftrightarrow \quad k^{\frac{1}{2}(p-1)} \equiv 1 \text{ or } -1 \pmod{m}.$$

The truth of this theorem follows immediately from the fact that, after any primitive root has been accepted as base, the index of the power $k^{\frac{1}{2}(p-1)}$ becomes either congruent to 0 or to $\frac{1}{2}(p-1)$, according as the index of k is even or odd.

Art. 57

For a given modulus it is easy enough to divide a complete system of incongruent residues into two classes, namely quadratic residues and non-residues; in this way all remaining numbers are assigned to their classes automatically. But the question of criteria distinguishing the moduli for which a given number is a quadratic residue from those for which it is a non-residue is much deeper.

As for the real units $+1$ and -1 , these are in fact squares in the arithmetic of complex numbers, and hence are quadratic residues for every modulus. Just as easily, from the criterion of the preceding article, it follows that the number i , and likewise $-i$, is a quadratic residue of every modulus whose norm p is of the form $8n + 1$, but a non-residue of every modulus whose norm is of the form $8n + 5$.

Since it plainly makes no difference whether the number m or one of the numbers associated with it, im , $-m$, or $-im$, is adopted as modulus, we may suppose that the modulus is the primary one among its associates (Article 36, II). Thus, putting the modulus equal to $a + bi$, a is odd and b is even. In this arrangement, since always $aa \equiv 1 \pmod{8}$, while bb is either $\equiv 0$

or $\equiv 4 \pmod{8}$ according as b is divisible by 4 or is even but not divisible by 4, it is clear that $+i$ and $-i$ are quadratic residues of the modulus in the former case and non-residues in the latter.

Art. 58

Since judging the character of a composite number, whether it is a quadratic residue or a non-residue, depends on the characters of its factors, it is plainly enough to limit the development of criteria distinguishing the moduli for which a given number k is a quadratic residue from those for which it is a non-residue to values of k that are primes and, moreover, primary among their associates.

In this investigation induction immediately supplies very elegant theorems. We begin with the number $1 + i$, which is found to be a quadratic residue of the moduli

$$-1 + 2i, +3 - 2i, -5 - 2i, -1 - 6i, +5 + 4i, +5 - 4i, -7, +7 + 2i, -5 + 6i, \dots$$

and a quadratic non-residue of the following:

$$-1 - 2i, -3, +3 + 2i, +1 + 4i, +1 - 4i, -5 + 2i, -1 + 6i, +7 - 2i, -5 - 6i, -3 + 8i, -3 - 8i, +5 + 8i, +5 - 8i, +9 + 4i, +9 - 4i, \dots$$

If we examine this list carefully, always taking from each set of four associated moduli the primary one, we readily observe that the moduli $a + bi$ in the first class are all those for which $a + b \equiv +1 \pmod{8}$, and those in the second are those for which $a + b \equiv -3 \pmod{8}$. Plainly, if instead of the primary modulus m we adopt its associate $-m$, this criterion must be changed so that for the moduli of the first class $a + b \equiv -1$, and for those of the second $a + b \equiv +3 \pmod{8}$.

Therefore, if induction has not deceived us, in general, for a prime number $a + bi$ in which a is odd and b is even, $1 + i$ becomes its quadratic residue or non-residue according as $a + b \equiv \pm 1$ or $\equiv \pm 3 \pmod{8}$.

For the number $-1 - i$ the same rule holds as for $1 + i$. On the other hand, regarding $1 - i$ as the product of $-i$ and $1 + i$, it is clear that $1 - i$ has the same character as $1 + i$ whenever b is divisible by 4, and the opposite character whenever b is even but not divisible by 4. Hence one easily obtains that $1 - i$ is a quadratic residue of the prime number $a + bi$ whenever $a - b \equiv \pm 1$, and a non-residue whenever $a - b \equiv \pm 3 \pmod{8}$, always supposing that a is odd and b even.

This second proposition may also be deduced from the first by means of the more general theorem stated as follows:

In the theory of quadratic residues, the character of the number $\alpha + \beta i$ with respect to the modulus $a + bi$ is the same as that of $\alpha - \beta i$ with respect to the modulus $a - bi$.

The proof of this theorem is drawn from the fact that the two moduli have the same norm p , and that whenever $(\alpha + \beta i)^{\frac{1}{2}(p-1)} - 1$ is divisible by $a + bi$, then $(\alpha - \beta i)^{\frac{1}{2}(p-1)} - 1$ also becomes divisible by $a - bi$; and whenever $(\alpha + \beta i)^{\frac{1}{2}(p-1)} + 1$ is divisible by $a + bi$, then $(\alpha - \beta i)^{\frac{1}{2}(p-1)} + 1$ must also be divisible by $a - bi$.

Art. 59

Let us proceed to odd prime numbers. We find that the number $-1 + 2i$ is a quadratic residue of the moduli

$$+3 + 2i, +1 - 4i, -5 + 2i, -5 - 2i, -1 - 6i, +7 - 2i, -3 + 8i, +5 + 8i, +5 - 8i, +9 + 4i, \dots$$

and a non-residue of the moduli

$$-1 - 2i, -3, +3 - 2i, +1 + 4i, -1 + 6i, +5 + 4i, +5 - 4i, -7, +7 + 2i, -5 + 6i, -5 - 6i, -3 - 8i, +9 - 4i, \dots$$

Reducing the moduli of the first class to their absolutely least residues modulo $-1 + 2i$, we find only $+1$ and -1 , for instance $+3 + 2i \equiv -1$, $+1 - 4i \equiv -1$, $-5 + 2i \equiv +1$, $-5 - 2i \equiv -1$, and so on. On the other hand, all moduli of the second class are found, modulo $-1 + 2i$, to be congruent either to $+i$ or to $-i$.

But the numbers $+1$ and -1 themselves are quadratic residues of the modulus $-1 + 2i$, while $+i$ and $-i$ are its non-residues. Therefore, so far as one may trust induction, the following theorem emerges: the number $-1 + 2i$ is a quadratic residue or non-residue of the prime number $a + bi$ according as that number is a quadratic residue or non-residue of $-1 + 2i$, provided that $a + bi$ is the primary one among its four associates, or rather, provided that a is odd and b even.

From this theorem there follow at once the analogous theorems concerning the numbers $+1 - 2i$, $-1 - 2i$, and $+1 + 2i$.

Art. 60

Carrying out a similar induction for the number -3 or $+3$, we find that both are quadratic residues of the moduli

$$+3 + 2i, +3 - 2i, -1 + 6i, -1 - 6i, -7, -5 + 6i, -5 - 6i, -3 + 8i, -3 - 8i, +9 + 4i, +9 - 4i, \dots$$

and non-residues of the following:

$$-1 + 2i, -1 - 2i, +1 + 4i, +1 - 4i, -5 + 2i, -5 - 2i, +5 + 4i, +5 - 4i, +7 + 2i, +7 - 2i, +5 + 8i, +5 - 8i, \dots$$

The former are, modulo 3, congruent to one of the four numbers $+1$, -1 , $+i$, $-i$; the latter to one of $+1 + i$, $+1 - i$, $-1 + i$, $-1 - i$. The former are precisely the quadratic residues of the number 3, the latter the non-residues.

This induction therefore teaches that a prime number $a + bi$, with a odd and b even, has with respect to -3 and also to $+3$ the same relation that -3 has to it, namely as regards whether one is the quadratic residue or non-residue of the other.

Extending a similar induction to other prime numbers, we find everywhere this very elegant law of reciprocity confirmed, and we are led to the following fundamental theorem concerning quadratic residues in the arithmetic of complex numbers:

If $a + bi$ and $A + Bi$ denote prime numbers such that a and A are odd, while b and B are even, then either each is a quadratic residue of the other, or each is a non-residue of the other.

Yet, despite the extreme simplicity of this theorem, its proof is beset with great difficulties. We shall not linger over them here, since the theorem itself is only a special case of a more general theorem that almost exhausts the whole theory of biquadratic residues. We now pass to that theory.

Art. 61

What was set forth in Article 2 of the preceding memoir concerning the notion of a biquadratic residue and non-residue we extend also to the arithmetic of complex numbers; and, as there, here too we restrict the inquiry to moduli that are prime numbers. At the same time it will usually be tacitly understood that the modulus is taken to be primary among its associates, for example congruent to 1 modulo $2 + 2i$, and also that the numbers whose character is under consideration, as biquadratic residues or non-residues, are not divisible by the modulus.

For a given modulus, therefore, the numbers not divisible by it could be divided into three classes: the first containing the biquadratic residues, the second the biquadratic non-residues that are nevertheless quadratic residues, and the third the quadratic non-residues.

But here too it is preferable, in place of the third class, to establish two classes, so that there are four in all. With any primitive root taken as base, the biquadratic residues will have indices divisible by 4, that is, of the form $4n$; the non-residues that are quadratic residues will have indices of the form $4n + 2$; finally, the indices of the quadratic non-residues will be partly of the form $4n + 1$ and partly of the form $4n + 3$. In this way four classes do arise, but the distinction between the last two would not be absolute; it would depend on the choice of the primitive root taken as base. For it is easy to see that one half of the primitive roots assign to a given quadratic non-residue an index of the form $4n + 1$, and the other half an index of the form $4n + 3$.

To remove this ambiguity, we shall always suppose such a primitive root to be adopted that the index $\frac{1}{4}(p-1)$ belongs to the number $+i$ (cf. Article 55, VI). This produces a classification that can more neatly be stated independently of primitive roots as follows. The first class contains those numbers k for which $k^{\frac{1}{4}(p-1)} \equiv 1$; these are the biquadratic residues of the modulus. The second class contains those for which $k^{\frac{1}{4}(p-1)} \equiv i$. The third contains those for which $k^{\frac{1}{4}(p-1)} \equiv -1$. Finally, the fourth contains those for which $k^{\frac{1}{4}(p-1)} \equiv -i$.

The third class comprises the biquadratic non-residues that are quadratic residues; the quadratic non-residues are distributed between the second and fourth. To the numbers of these classes we shall assign respectively the biquadratic characters 0, 1, 2, and 3. If we define the character λ of the number k modulo m to be the exponent of that power of i to which $k^{\frac{1}{4}(p-1)}$ is congruent, then plainly characters congruent modulo 4 are to be regarded as equivalent. For the present this notion is limited to prime moduli; in the continuation of these investigations we shall show how it can also be adapted to composite moduli.

Art. 62

In order that abundant induction concerning the characters of numbers may be built up more easily, we add here a compact table. With its aid the character of any proposed number with

respect to a modulus whose norm does not exceed 157 can be obtained with little work, provided the following observations are kept in mind.

Since the character of a composite number is equal, or congruent modulo 4, to the sum of the characters of its individual factors, it suffices to be able to assign, for a given modulus, the characters of prime numbers. Moreover, since the characters of the units -1 , i , and $-i$ are plainly congruent to the numbers $\frac{1}{2}(p-1)$, $\frac{1}{4}(p-1)$, and $\frac{3}{4}(p-1)$ modulo 4, it will also suffice to have displayed the characters of the numbers primary among their associates. Finally, since moduli congruent modulo m have the same character, it is enough to put into the table the characters of numbers contained in the system of absolutely least residues.

Furthermore, by reasoning similar to that of Article 58 one proves that, if for the modulus $a + bi$ the character of $A + Bi$ is λ , while for the modulus $a - bi$ the character of $A - Bi$ is λ' , then always $\lambda \equiv -\lambda' \pmod{4}$, or $\lambda + \lambda'$ is divisible by 4. Therefore it is enough to include in the table moduli for which b is either 0 or positive.

For example, if the character of $11 - 6i$ with respect to the modulus $-5 - 6i$ is sought, we substitute for these numbers $11 + 6i$ and $-5 + 6i$. We then determine (Article 43) the absolutely least residue of $11 + 6i$ modulo $-5 + 6i$, which is $-1 - 4i = -1 \times (1 + 4i)$. Since for the modulus $-5 + 6i$ the character of -1 is 30, while the character of $1 + 4i$ is, from the table, 2, the character of $11 + 6i$ modulo $-5 + 6i$ is 32, that is, 0; and hence, by the last observation, this is also the character of $11 - 6i$ modulo $-5 - 6i$.

Similarly, if the character of $-5 + 6i$ with respect to the modulus $11 + 6i$ is sought, its absolutely least residue $1 - 5i$ is resolved into the factors $-i$, $1 + i$, and $3 - 2i$, to which correspond the characters 117, 0, and 1. Hence the sought character is 118, that is, 2. The same character is also to be assigned to $-5 - 6i$ with respect to the modulus $11 - 6i$.

Modulus	Character	Numeri
-3	3	$1 + i$
$+3 + 2i$	3	$1 + i$
$+1 + 4i$	1	$-1 + 2i$
	3	$1 + i$
$-5 + 2i$	0	$-1 - 2i$
	1	$1 + i$
	2	$-1 + 2i$
$-1 + 6i$	0	-3
	1	$1 + i, -1 + 2i$
	2	$-1 - 2i$
$+5 + 4i$	0	$1 + i$
	1	-3
	3	$-1 + 2i, -1 - 2i$
-7	0	-3
	1	$-1 + 2i, 3 - 2i$
	2	$1 + i$
	3	$-1 - 2i, 3 + 2i$
$+7 + 2i$	0	$1 + i, 3 + 2i, 3 - 2i, 1 - 4i$
	1	-3

	2	$-1 - 2i, 1 + 4i$
	3	$-1 + 2i$
$-5 + 6i$	0	$1 + i, -3, 3 + 2i, 3 - 2i$
	1	$1 - 4i$
	2	$1 + 4i$
	3	$-1 + 2i, -1 - 2i$
$-3 + 8i$	0	$-1 + 2i, 3 - 2i, 1 - 4i$
	1	$1 + i, 3 + 2i$
	2	-3
	3	$-1 - 2i, 1 + 4i, -5 + 2i$
$+5 + 8i$	0	$-1 - 2i$
	1	$-5 - 2i, -1 + 6i$
	2	$-1 + 2i, 3 - 2i$
	3	$1 + i, -3, 3 + 2i, 1 + 4i, 1 - 4i$
$+9 + 4i$	0	$-1 + 2i, 3 + 2i$
	1	$1 + i, -1 - 2i, 3 - 2i$
	2	$-3, 1 + 4i$
	3	$1 - 4i, -5 + 2i$
$-1 + 10i$	0	$1 + i, -1 + 2i, -1 - 2i, 3 + 2i$
	1	-3
	2	$3 - 2i, -5 + 2i, 5 - 4i$
	3	$1 + 4i, 1 - 4i$
$+3 + 10i$	1	$1 + i, -1 - 2i, 1 - 4i$
	2	$-3, 3 + 2i, 1 + 4i, -5 - 2i$
	3	$-1 + 2i, 3 - 2i$
$-7 + 8i$	0	$1 + i, -7$
	1	$3 + 2i, 3 - 2i, 1 - 4i, -5 - 2i$
	2	$-1 - 2i, 1 + 4i, -5 + 2i, -1 - 6i$
	3	$-1 + 2i, -3, -1 + 6i$
-11	0	-3
	1	$1 + i, 3 - 2i, 1 + 4i, -5 + 2i, 5 + 4i$
	2	$-1 + 2i, -1 - 2i$
	3	$3 + 2i, 1 - 4i, -5 - 2i, 5 - 4i$
$-11 + 4i$	0	$1 + i, -1 + 2i, 3 + 2i, 5 + 4i$
	1	$-1 - 2i, -1 + 6i$
	2	$-5 + 2i$
	3	$-3, 3 - 2i, 1 + 4i, 1 - 4i, -5 - 2i$
$+7 + 10i$	0	$1 + 4i, 1 - 4i, -1 + 6i, -1 - 6i$
	1	$-1 + 2i, 3 + 2i, -5 + 2i$
	2	$1 + i, 3 - 2i$
	3	$-1 - 2i, -3, -5 - 2i$
$+11 + 6i$	0	$1 + i, -1 + 2i, -3, 1 + 4i, 1 - 4i, -7$
	1	$-1 - 2i, 3 + 2i, 3 - 2i$
	2	$-5 - 2i, -1 + 6i, 5 - 4i$
	3	$-5 + 2i, 5 + 4i, 7 - 2i$

Art. 63

We shall now take pains to detect by induction the common criteria for the moduli for which a given prime number has the same character. We always suppose the moduli to be primary among their associates, that is, numbers $a + bi$ for which either $a \equiv 1, b \equiv 0$, or $a \equiv 3, b \equiv 2 \pmod{4}$.

With respect to the number $1 + i$, from which we begin, the law of induction is more easily grasped if we separate the moduli of the first kind, for which $a \equiv 1, b \equiv 0$, from those of the second kind, for which $a \equiv 3, b \equiv 2$. With the aid of the table in the preceding article we find the following correspondence.

character	moduli of the first kind
0	$5 + 4i, -7 + 8i, -7 - 8i, -11 + 4i$
1	$1 - 4i, -3 + 8i, -3 - 8i, 9 + 4i, -11$
2	$5 - 4i, -7, -11 - 4i$
3	$-3, 1 + 4i, 5 + 8i, 5 - 8i, 9 - 4i$

If these seventeen examples are considered attentively, in all of them we find the character

$$\equiv \frac{1}{4}(a - b - 1) \pmod{4}.$$

Similarly there corresponds

character	moduli of the second kind
0	$3 - 2i, -1 - 6i, 7 + 2i, -5 + 6i, -1 + 10i, 11 + 6i$
1	$-5 + 2i, -1 + 6i, 7 - 2i, -1 - 10i, 3 + 10i$
2	$-1 + 2i, -5 - 2i, 3 - 10i, 7 + 10i$
3	$-1 - 2i, 3 + 2i, -5 - 6i, 7 - 10i, 11 - 6i$

In all these twenty examples, with slight attention, the character is found to be

$$\equiv \frac{1}{4}(a - b - 5) \pmod{4}.$$

These two rules can easily be contracted into one, valid for both kinds of moduli, if we note that $\frac{1}{4}bb$ is congruent to 0 for moduli of the first kind and to 1 for those of the second kind, modulo 4. Thus the character of $1 + i$ with respect to any prime modulus primary among its associates is

$$\equiv \frac{1}{4}(a - b - 1 - bb) \pmod{4}.$$

It is useful to note in passing that, since $(b + 1)^2$ is always of the form $8n + 1$, or $\frac{1}{4}(2b + bb)$ is even, this character is always even or odd according as $\frac{1}{4}(a + b - 1)$ is even or odd; this agrees with the rule for the quadratic character given in Article 58.

Since $\frac{1}{4}(a - b - 1)$ and $\frac{1}{4}(a - b + 3)$ are integers one of which is even and the other odd, their product will be even; that is,

$$\frac{1}{8}(a - b - 1)(a - b + 3) \equiv 0 \pmod{4}.$$

Therefore, in place of the expression just given for the biquadratic character, the following may also be adopted:

$$\frac{1}{4}(a - b - 1 - bb) - \frac{1}{8}(a - b - 1)(a - b + 3) = \frac{1}{8}(-aa + 2ab - 3bb + 1).$$

This form recommends itself also for the reason that it is not restricted to primary moduli, but only supposes a to be odd and b even. For under this supposition either $a + bi$ or $-a - bi$ will be primary among its associates, and the value of this formula is the same for both moduli.

Art. 64

Starting from the last rule found in the preceding article, we find that

number	character $\equiv$
$-1 + i$	$\frac{1}{8}(aa + 2ab - bb - 1)$
$-1 - i$	$\frac{1}{8}(-aa + 2ab + bb + 1)$
$+1 - i$	$\frac{1}{8}(aa + 2ab + 3bb - 1)$

This follows immediately from the fact that the character of i is $\frac{1}{4}(aa + bb - 1)$, while the character of -1 is $\frac{1}{2}(aa + bb - 1) \equiv \frac{1}{2}bb$, since $aa - 1$ is always of the form $8n$.

Plainly these four rules, although up to this point borrowed from induction, are so linked with one another that as soon as the proof of one has been completed, the other three will have been proved at the same time. It hardly needs to be said that in these rules also one supposes only that a is odd and b even.

If one does not dislike using formulas restricted to primary moduli, we may use this form:

number	character $\equiv$
$-1 + i$	$\frac{1}{4}(-a - b + 1 - bb)$
$-1 - i$	$\frac{1}{4}(a - b - 1 + bb)$
$+1 - i$	$\frac{1}{4}(-a - b + 1 + bb)$

The simplest formulas appear if, as we did at the beginning of our induction, we distinguish moduli of the first and second kind. Namely, the character is

number	for moduli of the first kind	for moduli of the second kind
$-1 + i$	$\frac{1}{4}(-a - b + 1)$	$\frac{1}{4}(-a - b - 3)$
$-1 - i$	$\frac{1}{4}(a - b - 1)$	$\frac{1}{4}(a - b + 3)$
$+1 - i$	$\frac{1}{4}(-a - b + 1)$	$\frac{1}{4}(-a - b + 5)$

Art. 65

For the number $-1 + 2i$, to which we now proceed, we shall use the same distinction between moduli $a + bi$ for which $a \equiv 1, b \equiv 0$, and those for which $a \equiv 3, b \equiv 2$. The table of Article 62 teaches that, with respect to this number, there corresponds

character	moduli of the first kind
0	$-3 + 8i, +5 - 8i, +9 + 4i, -11 + 4i$
1	$+1 + 4i, +5 - 4i, -7, -3 - 8i$
2	$+1 - 4i, +5 + 8i, -7 - 8i, -11$
3	$-3, +5 + 4i, +9 - 4i, -7 + 8i, -11 - 4i$

Reducing each of these moduli to its absolutely least residue modulo $-1 + 2i$, we notice that all those to which character 0 corresponds are congruent to 1; those to which character 1 corresponds are congruent to i ; those whose character is 2 become congruent to -1 ; finally, all those whose character is 3 become congruent to $-i$. But the characters of the numbers 1, i , -1 , and $-i$ modulo $-1 + 2i$ are respectively 0, 1, 2, and 3. Hence in all these seventeen examples the character of $-1 + 2i$ with respect to a modulus $a + bi$ of the first kind is identical with the character of that modulus with respect to the modulus $-1 + 2i$.

Similarly, with the aid of the table, there is found to correspond

character	moduli of the second kind
0	$+3 + 2i, -5 - 2i, -1 + 10i, -1 - 10i, +11 + 6i$
1	$+3 - 2i, -1 + 6i, -5 - 6i, +7 + 10i, +7 - 10i$
2	$-5 + 2i, -1 - 6i, +7 - 2i$
3	$-1 - 2i, +7 + 2i, -5 + 6i, +3 + 10i, +3 - 10i, +11 - 6i$

Reducing these moduli to least residues modulo $-1 + 2i$, all those to which the characters 0, 1, 2, and 3 respectively correspond are found congruent to the numbers -1 , $-i$, $+1$, and $+i$; but to these very numbers, if conversely $-1 + 2i$ is adopted as modulus, belong the characters 2, 3, 0, and 1, respectively. Therefore, in all these nineteen examples, the character of $-1 + 2i$ with respect to a modulus of the second kind differs by two units from the character of that modulus with respect to the number $-1 + 2i$ taken as modulus.

It is moreover seen without any difficulty that entirely similar facts will hold with respect to the number $-1 - 2i$.

Art. 66

For the number -3 we omit the distinction between moduli of the first and of the second kind, since the result shows that here it is superfluous. Thus there corresponds

character	moduli
0	$-1 + 6i, -1 - 6i, -7, -5 + 6i, -5 - 6i, -11, 11 + 6i, 11 - 6i$
1	$-1 - 2i, 1 - 4i, -5 + 2i, 5 + 4i, 7 + 2i, 5 - 8i, -1 + 10i, -7 - 8i, -11 - 4i, 7 - 10i$
2	$3 + 2i, 3 - 2i, -3 + 8i, -3 - 8i, 9 + 4i, 3 + 10i, 3 - 10i$
3	$-1 + 2i, 1 + 4i, -5 - 2i, 5 - 4i, 7 - 2i, 5 + 8i, -1 - 10i, -7 + 8i, -11 + 4i, 7 + 10i$

Reducing these moduli to least residues modulo 3, we see that those to which character 0 corresponds are partly $\equiv 1$ and partly $\equiv -1$; those whose character is 1 become either $\equiv 1 - i$

or $\equiv -1 + i$; those whose character is 2 become either $\equiv i$ or $\equiv -i$; finally, those to which character 3 belongs are either $\equiv 1 + i$ or $\equiv -1 - i$. From this induction we therefore infer that the character of the number -3 with respect to a modulus which is a primary prime among its associates is identical with the character of this same number when 3, or, what comes to the same thing, -3 , is considered as modulus.

Art. 67

By a similar induction carried out about other prime numbers, we find that the numbers $3 \pm 2i$, $-1 \pm 6i$, $7 \pm 2i$, $-5 \pm 6i$, etc., furnish theorems similar to the one at which we arrived in Article 65 with respect to the number $-1 + 2i$; on the other hand, the numbers $1 \pm 4i$, $5 \pm 4i$, $-3 \pm 8i$, $9 \pm 4i$, etc., behave in the same way as the number -3 .

Induction therefore leads to an exceedingly elegant theorem, which, after the model of the theory of quadratic residues in the arithmetic of real numbers, may be called the *fundamental theorem of the theory of biquadratic residues*, namely:

Let $a + bi$ and $a' + b'i$ denote distinct prime numbers, each primary among its associates, that is, congruent to unity modulo $2 + 2i$. The biquadratic character of the number $a + bi$ with respect to the modulus $a' + b'i$ will be identical with the character of the number $a' + b'i$ with respect to the modulus $a + bi$ if either both of the numbers $a + bi$, $a' + b'i$, or at least one of them, belongs to the first kind, that is, is congruent to unity modulo 4. On the other hand, those characters will differ from one another by two units if neither of the numbers $a + bi$, $a' + b'i$ belongs to the first kind, that is, if each is congruent to $3 + 2i$ modulo 4.

Notwithstanding the great simplicity of this theorem, however, its proof must be counted among the most deeply hidden mysteries of higher arithmetic, so that, at least as matters now stand, it can be unravelled only by very subtle investigations, which would far exceed the limits of the present memoir. We therefore reserve for the third memoir the publication of this proof, as well as the development of the connection between this theorem and the results which at the beginning of this memoir we had begun to establish by induction. As a kind of crown to the present discussion, however, we shall give here what is needed for the proof of the theorems proposed in Articles 63 and 64.

Art. 68

We begin with prime numbers $a + bi$ such that $b = 0$ (the third species of Article 34). Thus, in order that the number be primary among its associates, a must be a negative real prime number of the form $-(4n + 3)$; for it we shall write $-q$, such as -3 , -7 , -11 , -19 , and so on.

Let λ denote the character of the number $1 + i$ when that number is taken as modulus. We must have

$$i^\lambda \equiv (1 + i)^{\frac{1}{4}(qq-1)} \equiv 2^{\frac{1}{8}(qq-1)} \cdot i^{\frac{1}{8}(qq-1)} \pmod{q}.$$

But it is known that 2 is a quadratic residue, or a quadratic non-residue, of q according as q is of the form $8n + 7$, or of the form $8n + 3$. Hence we infer that, in general,

$$2^{\frac{1}{2}(q-1)} \equiv (-1)^{\frac{1}{4}(q+1)} \equiv i^{\frac{1}{2}(q+1)} \pmod{q},$$

and therefore, raising to the power whose exponent is $\frac{1}{4}(q+1)$,

$$2^{\frac{1}{8}(qq-1)} \equiv i^{\frac{1}{8}(q+1)^2} \pmod{q}.$$

The preceding equation therefore takes the form

$$i^\lambda \equiv i^{\frac{1}{8}(q+1)^2 + \frac{1}{8}(qq-1)} \equiv i^{\frac{1}{4}(qq+q)} \pmod{q},$$

from which it follows that

$$\lambda \equiv \frac{1}{4}(qq+q) \equiv \frac{1}{4}(q+1)^2 - \frac{1}{4}(q+1) \pmod{4},$$

or, since $\frac{1}{4}(q+1)^2 \equiv 0 \pmod{4}$,

$$\lambda \equiv -\frac{1}{4}(q+1) \equiv \frac{1}{4}(a-1) \pmod{4}.$$

This is precisely the theorem of Article 63 for the case $b = 0$.

Art. 69

Far more difficult are those moduli $a + bi$ for which b is not 0 (numbers of the fourth species in Article 34), and several investigations must be put in place first. We shall denote by p the norm $aa + bb$, which will be a real prime number of the form $4n + 1$.

Let S denote the complex of all simply least residues modulo $a + bi = m$, zero excluded; thus the number of entries contained in S is $p - 1$. Let $x + yi$ denote, indefinitely, a number of this system, and put

$$ax + by = \xi, \quad ay - bx = \eta.$$

Then ξ and η will be integers contained exclusively between the limits 0 and p . In the present case, where a and b are relatively prime, the formulas of Article 45, namely

$$\eta \equiv k\xi, \quad \xi \equiv -k\eta \pmod{p},$$

show that neither of the numbers ξ, η can be 0, unless the other vanishes at the same time, and hence $x = 0, y = 0$, a combination that we have already excluded. Thus the criterion for a number $x + yi$ contained in S consists in the positivity of the four numbers $\xi, \eta, p - \xi, p - \eta$.

We further observe that for no such number can $\xi = \eta$ hold; for then it would follow that

$$p(x + y) = a(\xi + \eta) + b(\xi - \eta) = 2a\xi,$$

which is absurd, since none of the factors 2, a, ξ is divisible by p . In a similar way the equation

$$p(x - y + a + b) = 2a\xi + (a + b)(p - \xi - \eta)$$

shows that $\xi + \eta = p$ cannot hold.

Therefore, since the numbers $\xi - \eta$ and $p - \xi - \eta$ must be either positive or negative, we derive from this a subdivision of the system S into four complexes C, C', C'', C''' , namely by putting into them as follows:

into the complex	the numbers for which
C	$\xi - \eta$ is positive, $p - \xi - \eta$ is positive
C'	$\xi - \eta$ is positive, $p - \xi - \eta$ is negative
C''	$\xi - \eta$ is negative, $p - \xi - \eta$ is negative
C'''	$\xi - \eta$ is negative, $p - \xi - \eta$ is positive

Thus the criterion for a number of the complex C is properly sixfold: the six numbers $\xi, \eta, p - \xi, p - \eta, \xi - \eta, p - \xi - \eta$ must be positive. But plainly conditions 2, 5, and 6 already imply the rest. Similar statements hold for the complexes C', C'', C''' , so that the complete criteria are triple:

for the complex	the numbers that must be positive
C	$\eta, \xi - \eta, p - \xi - \eta$
C'	$p - \xi, \xi - \eta, \xi + \eta - p$
C''	$p - \eta, \eta - \xi, \xi + \eta - p$
C'''	$\xi, \eta - \xi, p - \xi - \eta$

Moreover, even without our saying so, anyone will readily understand that, in the figurative representation of complex numbers (see Article 39), the numbers of the system S are contained inside the square whose sides join the points representing the numbers $0, a + bi, (1 + i)(a + bi)$, and $i(a + bi)$, and that the subdivision of the system S corresponds to the partition of the square by its diagonal lines. But at this point we have preferred to use purely arithmetical reasonings, leaving the illustration through geometric intuition to the experienced reader for the sake of brevity.

Art. 70

If four complex numbers $r = x + yi, r' = x' + y'i, r'' = x'' + y''i, r''' = x''' + y'''i$ are connected with one another so that

$$r' = m + ir, \quad r'' = m + ir' = (1 + i)m - r, \quad r''' = m + ir'' = im - ir,$$

and the first, r , is supposed to belong to the complex C , then the remaining numbers r', r'', r''' will belong respectively to the complexes C', C'', C''' .

Indeed, putting $\xi = ax + by, \eta = ay - bx, \xi' = ax' + by', \eta' = ay' - bx', \xi'' = ax'' + by'', \eta'' = ay'' - bx'', \xi''' = ax''' + by''', \eta''' = ay''' - bx'''$, one finds

$$\begin{aligned} \eta &= p - \xi' = p - \eta'' = \xi''', \\ \xi - \eta &= \xi' + \eta' - p = \eta'' - \xi'' = p - \xi''' - \eta''', \\ p - \xi - \eta &= \xi' - \eta' = \xi'' + \eta'' - p = \eta''' - \xi'''. \end{aligned}$$

The truth of the theorem then follows at once with the aid of the criteria.

And since again $r = m + ir'''$, it is easily seen that if r is supposed to belong to C' , then the numbers r', r'', r''' belong respectively to C'', C''', C ; if r belongs to C'' , they belong to C''', C, C' ; and finally, if r belongs to C''' , they belong to C, C', C'' . At the same time it follows that in each of the complexes C, C', C'', C''' there are equally many numbers, namely $\frac{1}{4}(p - 1)$.

Art. 71

Theorem. Let k denote an integer not divisible by m . Multiply each number of the complex C by k , and distribute the simply least residues of the products modulo m among the complexes C, C', C'', C''' . If the numbers of those residues which belong to the four complexes are denoted respectively by c, c', c'', c''' , then the character of the number k with respect to the modulus m will be

$$\equiv c' + 2c'' + 3c''' \pmod{4}.$$

Proof. Let the c least residues belonging to C be $\alpha, \beta, \gamma, \delta$, etc.; then let the c' residues belonging to C' be $m + i\alpha', m + i\beta', m + i\gamma', m + i\delta'$, etc.; further, let the c'' residues belonging to C'' be $(1+i)m - \alpha'', (1+i)m - \beta'', (1+i)m - \gamma'', (1+i)m - \delta''$, etc.; and finally let the c''' residues belonging to C''' be $im - i\alpha''', im - i\beta''', im - i\gamma''', im - i\delta'''$, etc.

Now consider four products, namely:

1. the product of all the $\frac{1}{4}(p-1)$ numbers that constitute the complex C ;
2. the product of the products obtained by multiplying each of these numbers by k ;
3. the product of the least residues of these products, namely of the numbers $\alpha, \beta, \gamma, \delta$, etc., $m + i\alpha', m + i\beta'$, etc. etc.;
4. the product of all the $c + c' + c'' + c'''$ numbers $\alpha, \beta, \gamma, \delta$, etc., α', β', γ' , etc., $\alpha'', \beta'', \gamma''$, etc., $\alpha''', \beta''', \gamma'''$, etc.

Denote these four products, in order, by P, P', P'', P''' . Plainly

$$P' = k^{\frac{1}{4}(p-1)} P, \quad P' \equiv P'', \quad P'' \equiv P''' i^{c' + 2c'' + 3c'''} \pmod{m},$$

and hence

$$Pk^{\frac{1}{4}(p-1)} \equiv P''' i^{c' + 2c'' + 3c'''} \pmod{m}.$$

But it is easily seen that the numbers α', β', γ' , etc., $\alpha'', \beta'', \gamma''$, etc., $\alpha''', \beta''', \gamma'''$, etc. all belong to the complex C , and that they are distinct both from one another and from the numbers $\alpha, \beta, \gamma, \delta$, etc., just as these latter are mutually distinct. All these numbers taken together, and abstracting from their order, must therefore be completely identical with all the numbers constituting the complex C . Hence we infer that $P = P'''$, and so

$$Pk^{\frac{1}{4}(p-1)} \equiv P i^{c' + 2c'' + 3c'''} \pmod{m}.$$

Finally, since none of the individual factors of the product P is divisible by m , it follows that

$$k^{\frac{1}{4}(p-1)} \equiv i^{c' + 2c'' + 3c'''} \pmod{m},$$

and hence $c' + 2c'' + 3c'''$ is the character of the number k with respect to the modulus m . Q. E. D.

Art. 72

In order that the general theorem of the preceding article may be applied to the number $1 + i$, the complex C must again be subdivided into two smaller complexes G and G' . We shall refer to G those numbers $x + yi$ for which $ax + by = \xi$ is less than $\frac{1}{2}p$, and to G' those for which ξ is greater than $\frac{1}{2}p$. We shall denote by g and g' respectively the number of entries contained in G and G' , whence $g + g' = \frac{1}{4}(p - 1)$.

The complete criterion for numbers belonging to G will therefore be that the three numbers η , $\xi - \eta$, $p - 2\xi$ are positive; for the third condition for the complex C , according to which $p - \xi - \eta$ must be positive, is already implicitly contained in these, since

$$p - \xi - \eta = (\xi - \eta) + (p - 2\xi).$$

Similarly, the complete criterion for numbers belonging to G' consists in positive values of the three numbers η , $p - \xi - \eta$, and $2\xi - p$.

It follows easily that the product of any number of the complex G by the number $1 + i$ belongs to the complex C''' . For if we put

$$\begin{aligned} (x + yi)(1 + i) &= x' + y'i, & ax' + by' &= \xi', & ay' - bx' &= \eta', \\ \text{one finds} \quad \xi' &= \xi - \eta, & \eta' - \xi' &= 2\eta, & p - \xi' - \eta' &= p - 2\xi, \end{aligned}$$

that is, the criterion for a number $x + yi$ placed in the complex G is identical with the criterion for the number $x' + y'i$ belonging to the complex C''' . In exactly the same way it is shown that the product of any number of G' by $1 + i$ belongs to the complex C''' .

Therefore, if in the preceding article we give k the value $1 + i$, then $c = 0$, $c' = 0$, $c'' = g'$, $c''' = g$, and hence the character of the number $1 + i$ becomes

$$3g + 2g' = \frac{1}{2}(p - 1) + g.$$

And since the characters of the numbers i and -1 are $\frac{1}{4}(p - 1)$ and $\frac{1}{2}(p - 1)$, respectively, the characters of the numbers $-1 + i$, $-1 - i$, $1 - i$ are respectively

$$\frac{3}{4}(p - 1) + g, \quad g, \quad \frac{1}{4}(p - 1) + g.$$

Thus the whole crux of the matter now turns on the investigation of the number g .

Art. 73

What we have set out in Articles 69–72 is properly independent of the supposition that m is a primary number. From now on, however, we shall at least suppose that a is odd, b even, and moreover that a , b , and $a - b$ are positive numbers.

Above all, the limits of the values of x in the complex G must be established. Putting

$$ay - bx = \eta, \quad (a + b)x - (a - b)y = \zeta, \quad p - 2ax - 2by = \theta,$$

the criterion for numbers $x + yi$ belonging to the complex G consists in the three conditions that η , ζ , and θ are positive numbers.

Since

$$px = (a - b)\eta + a\zeta, \quad p(a - 2x) = a\theta + 2b\eta,$$

it is plain that x and $2a - x$ must be positive numbers; that is, x must be equal to one of the numbers $1, 2, 3, \dots, \frac{1}{2}(a - 1)$.

Further, since

$$(a - b)\theta = 2b\zeta + p(a - b - 2x),$$

it is clear that as long as x is less than $\frac{1}{2}(a - b)$, the second condition (according to which ζ must be positive) already implies the third (that θ must be positive); on the other hand, whenever x is greater than $\frac{1}{2}(a - b)$, the second condition is already contained under the third. Therefore, for the values of x equal to $1, 2, 3, \dots, \frac{1}{2}(a - b - 1)$, one need only ensure that η and ζ become positive; that is, that y is greater than $\frac{bx}{a}$ and less than $\frac{(a+b)x}{a-b}$. Thus, for such a given value of x , there will be exactly

$$\left[\frac{(a+b)x}{a-b} \right] - \left[\frac{bx}{a} \right]$$

numbers $x + yi$, if we use brackets in the same meaning in which we have already used them in many other places.

On the other hand, for the values of x equal to $\frac{1}{2}(a - b + 1), \frac{1}{2}(a - b + 3), \dots, \frac{1}{2}(a - 1)$, it will suffice to make η and θ positive; that is, y must be greater than $\frac{bx}{a}$ and less than

$$\frac{p - 2ax}{2b} = \frac{1}{2}b + \frac{aa - 2ax}{2b}.$$

Hence, for such a given value of x , there will be exactly

$$\left[\frac{1}{2}b + \frac{aa - 2ax}{2b} \right] - \left[\frac{bx}{a} \right]$$

numbers $x + yi$. We therefore infer that the number of entries of the complex G is

$$g = \Sigma \left[\frac{(a+b)x}{a-b} \right] + \Sigma \left[\frac{1}{2}b + \frac{aa - 2ax}{2b} \right] - \Sigma \left[\frac{bx}{a} \right],$$

where in the first term the summation is to be extended over all integral values of x from 1 to $\frac{1}{2}(a - b - 1)$; in the second from $\frac{1}{2}(a - b + 1)$ to $\frac{1}{2}(a - 1)$; in the third from 1 to $\frac{1}{2}(a - 1)$.

If we use the characteristic φ in the same meaning as in the cited place, namely

$$\varphi(t, u) = \left[\frac{u}{t} \right] + \left[\frac{2u}{t} \right] + \left[\frac{3u}{t} \right] + \dots + \left[\frac{t'u}{t} \right],$$

where t and u denote any positive numbers and t' denotes the number $\left[\frac{1}{2}t \right]$, then the first term becomes $\varphi(a - b, a + b)$, and the third becomes $-\varphi(a, b)$. The second becomes

$$\frac{1}{4}bb + \Sigma \left[\frac{aa - 2ax}{2b} \right].$$

But, writing the terms in the reverse order,

$$\Sigma \left[\frac{aa - 2ax}{2b} \right] = \left[\frac{a}{2b} \right] + \left[\frac{3a}{2b} \right] + \left[\frac{5a}{2b} \right] + \cdots + \left[\frac{(b-1)a}{2b} \right] = \varphi(2b, a) - \varphi(b, a).$$

Thus our formula takes the form

$$g = \varphi(a - b, a + b) + \varphi(2b, a) - \varphi(a, b) - \varphi(b, a) + \frac{1}{4}bb.$$

First consider the term $\varphi(a - b, a + b)$, which is at once transformed into

$$\varphi(a - b, 2b) + 1 + 2 + 3 + \cdots + \frac{1}{2}(a - b - 1),$$

or into

$$\varphi(a - b, 2b) + \frac{1}{8}((a - b)^2 - 1).$$

Next, since by the general theorem

$$\varphi(t, u) + \varphi(u, t) = \left[\frac{1}{2}t \right] \cdot \left[\frac{1}{2}u \right],$$

when t and u are positive relatively prime integers, we have

$$\varphi(a - b, 2b) = \frac{1}{2}b(a - b - 1) - \varphi(2b, a - b),$$

and therefore

$$\varphi(a - b, a + b) = \frac{1}{8}(aa + 2ab - 3bb - 4b - 1) - \varphi(2b, a - b).$$

Let us arrange the parts of $\varphi(2b, a - b)$ as follows:

$$\begin{aligned} & \left[\frac{a - b}{2b} \right] + \left[\frac{3(a - b)}{2b} \right] + \left[\frac{5(a - b)}{2b} \right] + \cdots + \left[\frac{(b - 1)(a - b)}{2b} \right] \\ & + \left[\frac{a - b}{b} \right] + \left[\frac{2(a - b)}{b} \right] + \left[\frac{3(a - b)}{b} \right] + \cdots + \left[\frac{\frac{1}{2}b(a - b)}{b} \right]. \end{aligned}$$

The second series plainly becomes

$$\varphi(b, a - b) = \varphi(b, a) - 1 - 2 - 3 - \cdots - \frac{1}{2}b = \varphi(b, a) - \frac{1}{8}(bb + 2b).$$

We write the first series in the reverse order of its terms as

$$\begin{aligned} & \left[\frac{1}{2}(a + 1 - b) - \frac{a}{2b} \right] + \left[\frac{1}{2}(a + 3 - b) - \frac{3a}{2b} \right] \\ & + \left[\frac{1}{2}(a + 5 - b) - \frac{5a}{2b} \right] + \cdots + \left[\frac{1}{2}(a - 1) - \frac{(b - 1)a}{2b} \right], \end{aligned}$$

which expression, since in general, if t denotes an integer and u a fraction, $[t - u] = t - 1 - [u]$, is changed into

$$\begin{aligned} & \frac{1}{8}b(2a - 4 - b) - \left[\frac{a}{2b}\right] - \left[\frac{3a}{2b}\right] - \left[\frac{5a}{2b}\right] - \dots - \left[\frac{(b-1)a}{2b}\right] \\ &= \frac{1}{8}b(2a - 4 - b) - \varphi(2b, a) + \varphi(b, a). \end{aligned}$$

Hence

$$\varphi(2b, a - b) = 2\varphi(b, a) - \varphi(2b, a) + \frac{1}{4}b(a - 3 - b),$$

and consequently

$$\varphi(a - b, a + b) = \varphi(2b, a) - 2\varphi(b, a) + \frac{1}{8}(aa - bb + 2b - 1).$$

Substituting this value in the formula for g given above, and also using

$$\varphi(a, b) + \varphi(b, a) = \frac{1}{4}b(a - 1),$$

we obtain

$$g = 2\varphi(2b, a) - 2\varphi(b, a) + \frac{1}{8}(aa - 2ab + bb + 4b - 1).$$

Art. 74

By entirely similar reasonings one handles the case in which a and b remain positive but $a - b$ is negative, or $b - a$ positive. The equations

$$p(a - 2x) = 2b\eta + a\theta, \quad p(b - a + 2x) = 2b\zeta + (b - a)\theta$$

show that $\frac{1}{2}a - x$ and $x + \frac{1}{2}(b - a)$ are positive, and hence that x must be equal to one of the numbers

$$-\frac{1}{2}(b - a - 1), -\frac{1}{2}(b - a - 3), -\frac{1}{2}(b - a - 5), \dots, +\frac{1}{2}(a - 1).$$

Further, from the equation $px + (b - a)\eta = a\zeta$ it follows that, for negative values of x , the condition by which η must be positive is already contained under the condition by which ζ must be positive; the opposite happens whenever a positive value is assigned to x . Hence the values of y , for a fixed negative value of x , must be contained between $\frac{(a+b)x}{a-b}$ and $\frac{p-2ax}{2b}$, while for a positive value of x they must be contained between $\frac{bx}{a}$ and $\frac{p-2ax}{2b}$. Plainly for $x = 0$ these limits are 0 and $\frac{p-2ax}{2b}$, the value $y = 0$ itself being excluded.

It follows that

$$g = - \sum \left[\frac{(a+b)x}{a-b} \right] + \sum \left[\frac{1}{2}b + \frac{aa - 2ax}{2b} \right] - \sum \left[\frac{bx}{a} \right],$$

where in the first term the summation is to be extended over all negative values of x from -1 to $-\frac{1}{2}(b - a - 1)$; in the second over all values of x from $-\frac{1}{2}(b - a - 1)$ to $\frac{1}{2}(a - 1)$; in the

third over all positive values of x from $+1$ to $\frac{1}{2}(a-1)$. In this way the first summation gives $-\varphi(b-a, b+a)$; the second, as in the preceding article, gives $\frac{1}{4}bb + \varphi(2b, a) - \varphi(b, a)$; and the third gives $-\varphi(a, b)$. Thus

$$g = -\varphi(b-a, b+a) + \varphi(2b, a) - \varphi(b, a) - \varphi(a, b) + \frac{1}{4}bb.$$

Now, in the same way as in the preceding article,

$$\begin{aligned}\varphi(b-a, b+a) &= \varphi(b-a, 2b) - \frac{1}{8}((b-a)^2 - 1) \\ &= \frac{1}{8}(3bb - 2ab - aa - 4b + 1) - \varphi(2b, b-a),\end{aligned}$$

and also

$$\varphi(2b, b-a) = \varphi(2b, a) - 2\varphi(b, a) + \frac{1}{4}b(b-1-a),$$

so that

$$\varphi(b-a, b+a) = 2\varphi(b, a) - \varphi(2b, a) + \frac{1}{8}(bb - aa - 2b + 1),$$

and finally

$$g = 2\varphi(2b, a) - 2\varphi(b, a) + \frac{1}{8}(aa - 2ab + bb + 4b - 1).$$

It has therefore been established that the same formula for g holds whether $a-b$ is positive or negative, provided a and b are positive.

Art. 75

To obtain the further reduction, put

$$\begin{aligned}L &= \left[\frac{a}{2b} \right] + \left[\frac{2a}{2b} \right] + \left[\frac{3a}{2b} \right] + \cdots + \left[\frac{\frac{1}{2}ba}{2b} \right], \\ M &= \left[\frac{(\frac{1}{2}b+1)a}{2b} \right] + \left[\frac{(\frac{1}{2}b+2)a}{2b} \right] + \left[\frac{(\frac{1}{2}b+3)a}{2b} \right] + \cdots + \left[\frac{ba}{2b} \right], \\ N &= \left[\frac{a+b}{2b} \right] + \left[\frac{2a+b}{2b} \right] + \left[\frac{3a+b}{2b} \right] + \cdots + \left[\frac{\frac{1}{2}ba+b}{2b} \right].\end{aligned}$$

Since it is easily seen that, in general, $[u] + [u + \frac{1}{2}] = [2u]$ for any real quantity u , we get $L + N = \varphi(b, a)$; and since plainly $L + M = \varphi(2b, a)$, we have

$$\varphi(2b, a) - \varphi(b, a) = M - N.$$

Further, it is obvious that the sum of the first term of the series N with the penultimate term of the series M , namely

$$\left[\frac{a+b}{2b} \right] + \left[\frac{(b-1)a}{2b} \right],$$

becomes $\frac{1}{2}(a-1)$, and that the same sum is produced from the second term of the series N with the antepenultimate term of M , and so on. Hence, since also the last term of the series M becomes $\frac{1}{2}(a-1)$, while the last term of the series N is

$$\left[\frac{a+2}{4} \right] = \frac{1}{4}(a \mp 1),$$

with the upper or lower sign holding according as a is of the form $4n+1$ or $4n-1$, we get

$$M+N = \frac{1}{4}(a-1)b + \frac{1}{4}(a \mp 1),$$

and hence

$$\varphi(2b, a) - \varphi(b, a) = \frac{1}{4}(a-1)b + \frac{1}{4}(a \mp 1) - 2N.$$

Thus the formula for g found in Articles 73 and 74 passes into

$$g = \frac{1}{8}((a+b)^2 - 1) + 2n - 4N,$$

putting $a \mp 1 = 4n$, where n is an integer. But since from this we have $1 = 16nn - 8an + aa$, the formula may also be written in the following way:

$$g = \frac{1}{8}(-aa + 2ab + bb + 1) + 4 \left(\frac{1}{2}(a+1)n - nn - N \right).$$

Consequently, since g is the character of the number $-1-i$ with respect to the modulus $a+bi$, this character becomes

$$\equiv \frac{1}{8}(-aa + 2ab + bb + 1) \pmod{4}.$$

This is precisely the theorem found above by induction in Article 64, and from it the theorems about the characters of the numbers $1+i$, $1-i$, and $-1+i$ follow at once. Therefore these four theorems have now been rigorously proved for the case in which a and b are positive.

Art. 76

If, while a remains positive, b is negative, put $b = -b'$, so that b' is positive. Since it has now been proved that, for the modulus $a+b'i$, the character of the number $-1-i$ is

$$\equiv \frac{1}{8}(-aa + 2ab' + b'b' + 1) \pmod{4},$$

the character of the number $-1+i$ for the modulus $a-b'i$, by the theorem stated in Article 62, will be

$$\equiv \frac{1}{8}(aa - 2ab' - b'b' - 1).$$

That is, the character of the number $-1+i$ for the modulus $a+bi$ becomes

$$\equiv \frac{1}{8}(aa + 2ab - bb - 1).$$

But this is exactly the theorem stated in Article 64, from which the remaining three statements about the characters of $1 + i$, $1 - i$, and $-1 - i$ follow at once. Therefore these theorems have also been proved for the case in which b is negative, that is, for all cases in which a is positive.

Finally, if a is negative, put $a = -a'$, $b = -b'$. Since by what has already been proved the character of the number $1 + i$ with respect to the modulus $a' + b'i$ is

$$\equiv \frac{1}{8}(-a'a' + 2a'b' - 3b'b' + 1) \pmod{4},$$

and since it makes no difference whether we take $a' + b'i$ or its opposite $-a' - b'i$ as modulus, it is plain that the character of the number $1 + i$ with respect to the modulus $a + bi$ is

$$\equiv \frac{1}{8}(-aa + 2ab - 3bb + 1),$$

and similar statements hold about the characters of the numbers $1 - i$, $-1 + i$, and $-1 - i$. It follows from this that the proof of the theorems about the characters of the numbers $1 + i$, $1 - i$, $-1 + i$, and $-1 - i$ (Articles 63 and 64) is no longer subject to any limitation.